

EDUCATIVE COMMENTARY ON ISI 2014 MATHEMATICS PAPERS

(Last updated on 23/08/2017)

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Introduction

This is a **revised** version of that loaded about a month ago. The numerical slip in the draft version which put a question mark about Q.29 in Paper 1 was sorted out by S. Venkatesh. The subsequently added alternative solution to Q.1 in Paper 1 was contributed by Vinayak Antarkar while that to Q.23 was contributed by Deepanshu Rajvanshi. Siddhesh Naik pointed out that the assertion made in the solution of Q.17 to the effect that the graphs of $y = f(x)$ and $y = f^{-1}(x)$ intersect only on the line $y = x$ is not correct. The solution given remains valid but there is no sneaky path as claimed earlier.

The shorter solutions to Q.5 and Q.27 in Paper 1 were also contributed by Siddhesh Naik. He further pointed out that the use of the structure of the Pythagorean triples is not vital to arrive at the answer to Q.15 in Paper 1. The solution to Q.26 using the triangle inequality was supplied by Vinayak Antarkar.

The only significant addition in the present version is the comment at the end of Q.27 in Paper 1 that it is a straight copy of a 2003 question in the Putnam competition. This was pointed out by a reader who probably prefers to remain anonymous.

As in the commentary on ISI 2016 papers, all the references in the present commentary will be to *Educative JEE Mathematics* by the author, unless

otherwise stated. The pagination will be as per its third edition, which is available in the market. Also from time to time, comparison will be made with JEE which is a far more well known examination.

As Paper 1 consists entirely of Multiple Choice Questions, the acronym MCQ will appear frequently in the comments.

ISI BStat-BMath-UGA-2014 Paper with Comments

N.B. Each question has four options of which ONLY ONE is correct.

Q.1 The system of inequalities

$$a - b^2 \geq \frac{1}{4}, b - c^2 \geq \frac{1}{4}, c - d^2 \geq \frac{1}{4}, d - a^2 \geq \frac{1}{4} \text{ has}$$

- (A) no solutions (B) exactly one solution
(C) exactly two solutions (D) infinitely many solutions.

Answer and Comments: (B). Normally, an inequality of real numbers has infinitely many solutions. But in degenerate cases, the number of solutions may reduce to a finite number or even 0. For example, if $c > 0$ then the inequality $x^2 + y^2 \leq c$ is satisfied by all points inside and on the circle of radius $\sqrt{c}$ centred at the origin. But if $c = 0$, there is only one solution, $x = 0, y = 0$. And for $c < 0$ there are no (real) solutions. The same thing holds for systems of inequalities. Their solution sets are the intersections of the solutions sets of the individual inequalities and sometimes the intersection of two infinite sets can be a finite set, as for example, the intersection of the sets $[a, \infty)$ and $(-\infty, a]$ which are, respectively, the solution sets of the inequalities $x \geq a$ and $x \leq a$.

In the present problem we have a system of four inequalities. They are interrelated and in fact have a common form. It is not given that a, b, c, d are positive. But we see this from $a \geq b^2 + \frac{1}{4}$ etc. Now, by the A.M.-G.M. inequality, we have

$$a \geq b^2 + \frac{1}{4} \geq 2 \times b \times \frac{1}{2} = b \tag{1}$$

So, the first inequality implies

$$a \geq b \tag{2}$$

In a cyclic manner the remaining three inequalities imply that $b \geq c, c \geq d$ and $d \geq a$. Put together, we get that a, b, c, d are all equal.

This also means that equality has to hold in (1), which can happen only when $b^2 = \frac{1}{4}$, i.e. when $b = \frac{1}{2}$. So the system has only one solution in which every a, b, c, d equals $\frac{1}{2}$. ■

It is tempting to add the four inequalities to get

$$a + b + c + d \geq (a^2 + b^2 + c^2 + d^2) + 1 \quad (3)$$

and try to express $a^2 + b^2 + c^2 + d^2$ in terms of $(a + b + c + d)^2$. But then we have to subtract 6 terms of the form $2ab$ and the expression becomes complicated. It is also to be noted that (3) is weaker than the simultaneous holding of the four given inequalities. By diluting the hypothesis we are not likely to reach the solution. But in the present case, we can subtract $a + b + c + d$ from both the sides and get

$$(a - \frac{1}{2})^2 + (b - \frac{1}{2})^2 + (c - \frac{1}{2})^2 + (d - \frac{1}{2})^2 \leq 0 \quad (4)$$

A sum of four squares can never be negative. And it equals 0 only when each one of them equals 0. This gives a slicker solution. Such methods work in critical cases where inequalities become equalities. If the R.H.S. of (4) were 1 (or some other positive number) we would not be able to conclude anything. (As pointed out in Chapter 14, many equations in trigonometry arising from optimisation in triangles fall in this category, for example, the equation $\cos A + \cos B + \cos C = \frac{3}{2}$.)

Although problems based on the A.M.-G.M. inequality are very common, the present one is a little novel in that several such inequalities are given together. Once the idea of using the A.M.-G.M. strikes, the computations involved are simple and do not take much time. So this problem is well-suited as an MCQ.

Q.2 Let $\log_{12} 18 = a$. Then $\log_{24} 16$ is equal to

$$(A) \frac{8-3a}{5-a} \quad (B) \frac{1}{3+a} \quad (C) \frac{4a-1}{2+3a} \quad (D) \frac{8-4a}{5+a}$$

Answer and Comments: (A). The problem involves logarithms taken with different bases. In such problems, they all have to be converted to

logarithms w.r.t. a common base, say b . The choice of b is arbitrary as long as it is positive. But it should not be 1 because logarithms with base 1 are not well defined. Also if we choose $b \in (0, 1)$, we have to be careful that larger numbers have smaller logarithms. This is especially important if the problem involves inequalities about logarithms. This complication does not arise in the present problem. But instead of taking some arbitrary base, we choose it in a way that some of the logarithms are simplified. In the present problem both the bases, viz. 12 and 24 are 3 times some powers of 2. So, it is better that we take b to be either 2 or 3. We choose 2 partly because powers of 2 are easier to identify than those of 3.

Next we express the given logarithm in terms of logarithms with base 2. The standard formula for this conversion is

$$\log_x y = \frac{\log_b y}{\log_b x} \quad (1)$$

So, in our problem,

$$a = \log_{12} 18 = \frac{\log_2 18}{\log_2 12} \quad (2)$$

We simplify the logarithms in the numerator as well as in the denominator using the basic rules of logarithms, viz.

$$\log_b xy = \log_b x + \log_b y \quad (3)$$

$$\text{and } \log_b(x^r) = r \log_b x \quad (4)$$

Since $18 = 3^2 \times 2$ and $12 = 2^2 \times 3$, repeated applications of these rules (with $b = 2$) and the fact that $\log_2 2 = 1$ give

$$a = \frac{2 \log_2 3 + \log_2 2}{2 \log_2 2 + \log_2 3} = \frac{2 \log_2 3 + 1}{2 + \log_2 3} \quad (5)$$

Thus we have expressed the given log in terms of $\log_2 3$. For notational simplicity, call this number u . Then (5) becomes

$$a = \frac{2u + 1}{2 + u} \quad (6)$$

which can be recast to express u in terms of a as

$$u = \frac{2a - 1}{2 - a} \quad (7)$$

So if we are able to express $\log_{24} 16$ in terms of u then we can express it in terms of a using (7). For this we use (1) again with $b = 2$ to get

$$\begin{aligned}\log_{24} 16 &= \frac{\log_2 16}{\log_2 24} = \frac{4}{\log_2 8 + \log_2 3} \\ &= \frac{4}{3 + u} = \frac{4}{3 + \frac{2a-1}{2-a}} = \frac{8-4a}{5-a}\end{aligned}\quad (8)$$

Therefore (A) is correct. ■

A simple problem based on standard properties of logarithms. But the computational work needed is a bit lengthy and prone to slips of hand.

Q.3 The number of solutions of the equation $\tan x + \sec x = 2 \cos x$, where $0 \leq x \leq \pi$, is

(A) 0 (B) 1 (C) 2 (D) 3.

Answer and Comments: (C). The paper setters should have been careful to exclude the point $\pi/2$ because at this point, neither $\tan x$ nor $\sec x$ is defined. For other values of $x \in [0, \pi]$ the given equation reduces to

$$\sin x + 1 = 2 \cos^2 x \quad (1)$$

which can be written as a quadratic in $\sin x$, viz.

$$2 \sin^2 x + \sin x - 1 = 0 \quad (2)$$

which gives

$$\sin x = \frac{-1 \pm 3}{4} \quad (3)$$

So $\sin x = \frac{1}{2}$ or $\sin x = -1$. The first equation has two solutions in $[0, \pi]$ (which are easy to identify as $\pi/6$ and $5\pi/6$ but that is not asked). The second equation has none. So the given equation has two solutions. ■

Generally, whenever we reduce a given equation to some other, all the solutions of the original equation are also of the second. The

converse may fail. So one has to check that the two solutions we got are indeed solutions of the given equation too. That is very easy here and most students do not bother about such points. In an MCQ they stand rewarded in terms of the time saved. Had the problem been designed so that some solutions of the new equation have to be discarded, it would have been a truly testing problem. As it stands, the problem is absolutely straightforward and simple.

Q.4 Using only the digits 2, 3 and 9, how many six digit numbers can be formed which are divisible by 6?

- (A) 41 (B) 80 (C) 81 (D) 161

Answer and Comments: (C). Let the number be $n = abcdef$. Each entry can take three possible values, 2, 3 and 9 independently. So in all there are 3^6 such numbers. But we want only those in which n is divisible by 6. This is equivalent to n being divisible by both 2 and 3. The first requirement forces f to be an even digit and hence only 2. The second requirement implies that the sum $a + b + c + d + e + f$ is divisible by 3. As 3 and 9 are each divisible by 3 and $f = 2$, there are only two possibilities for n to be divisible by 3, (i) exactly two of a, b, c, d and e are 2 and (ii) all five of them are 2. When (i) holds, the remaining three places can be filled by 3 and 9 in $2^3 = 8$ ways. So there are $\binom{5}{2} \times 8 = 80$ numbers of type (i). There is only one number, viz. 222222 of type (ii). So the total count is 81. ■

A nice simple counting problem requiring only very elementary number theory. The calculations are easy and can be done mentally. And there is no way to arrive at the answer sneakily. So an ideal question for an MCQ test.

Q.5 What is the value of the following integral?

$$\int_{1/2014}^{2014} \frac{\tan^{-1} x}{x} dx$$

- (A) $\frac{\pi}{4} \log 2014$ (B) $\frac{\pi}{2} \log 2014$ (C) $\pi \log 2014$ (D) $\frac{1}{2} \log 2014$

Answer and Comments: (B). The only role 2014 has in this problem is that it declares that it is a part of a test given in the year 2014. Call the integral as I . We cannot easily find an antiderivative of the integrand. So this integral has to be evaluated by some special method such as those given in Chapter 18. The general principle of these methods is to convert I to some other form, say J , so that $I + J$ can be evaluated relatively easily. We first use the substitution $x = \tan \theta$ to convert I . Denote $\tan^{-1}(2014)$ by α . Then $\tan^{-1}(\frac{1}{2014}) = \frac{\pi}{2} - \alpha$.

$$I = \int_{\frac{\pi}{2}-\alpha}^{\alpha} \frac{\theta}{\tan \theta} \sec^2 \theta d\theta = \int_{\frac{\pi}{2}-\alpha}^{\alpha} \frac{\theta}{\sin \theta \cos \theta} d\theta \quad (1)$$

Next, we use the ‘reflection principle’ (see p. 696) which says that

$$\int_a^b f(u) du = \int_a^b f(a+b-u) du \quad (2)$$

So,

$$\begin{aligned} I &= \int_{\frac{\pi}{2}-\alpha}^{\alpha} \frac{\frac{\pi}{2} - \theta}{\sin(\frac{\pi}{2} - \theta) \cos(\frac{\pi}{2} - \theta)} d\theta \\ &= \int_{\frac{\pi}{2}-\alpha}^{\alpha} \frac{\frac{\pi}{2} - \theta}{\sin \theta \cos \theta} d\theta = \int_{\frac{\pi}{2}-\alpha}^{\alpha} \frac{\frac{\pi}{2}}{\sin \theta \cos \theta} d\theta - I \end{aligned} \quad (3)$$

If we call this as J then $I + J = 2I$ which is easier to evaluate than I .

$$\begin{aligned} 2I &= \pi \int_{\frac{\pi}{2}-\alpha}^{\alpha} \frac{d\theta}{\sin 2\theta} = \pi \int_{\frac{\pi}{2}-\alpha}^{\alpha} \operatorname{cosec} 2\theta d\theta \\ &= -\frac{\pi}{2} \log(\cot 2\theta + \operatorname{cosec} 2\theta) \Big|_{\frac{\pi}{2}-\alpha}^{\alpha} \end{aligned} \quad (4)$$

This completes the integration part of the problem. The rest consists of simplification using trigonometry and properties of logarithms. Note that

$$\cot 2\theta + \operatorname{cosec} 2\theta = \frac{\cos 2\theta + 1}{\sin 2\theta} = \cot \theta \quad (5)$$

So, (4) continues as

$$\begin{aligned} 2I &= -\frac{\pi}{2} \log(\cot \theta) \Big|_{\frac{\pi}{2}-\alpha}^{\alpha} \\ &= \frac{\pi}{2} [\log(\tan \alpha) - \log(\cot \alpha)] \\ &= \frac{\pi}{2} \log(\tan^2 \alpha) = \pi \log \tan \alpha = \pi \log 2014 \end{aligned} \quad (6)$$

So, finally, $I = \frac{\pi}{2} \log 2014$. Hence (B) is correct. ■

A more tricky solution is to put $y = 1/x$ to convert I to $\int_{1/2014}^{2014} \frac{\frac{\pi}{2} - \tan^{-1} y}{y} dy$. Then $2I = \frac{\pi}{2} \int_{1/2014}^{2014} \frac{dy}{y} = \pi \log(2014)$.

Very similar problems have been asked in the JEE. See the long list in Exercise (18.15). So the tricky part of the problem is no longer so tricky if attempted by the first method. It is only the second solution which is based on an unusual trick. A faint hint for the substitution $y = \frac{1}{x}$ comes from the fact that the upper and the lower limits of integration are reciprocals of each other. Perhaps the paper setters intended to reward those who can take such hints. Others can also get the answer, but with a lot of work.

Q.6 A light ray travelling along the line $y = 1$, is reflected by a mirror placed along the line $x = 2y$. The reflected ray travels along the line

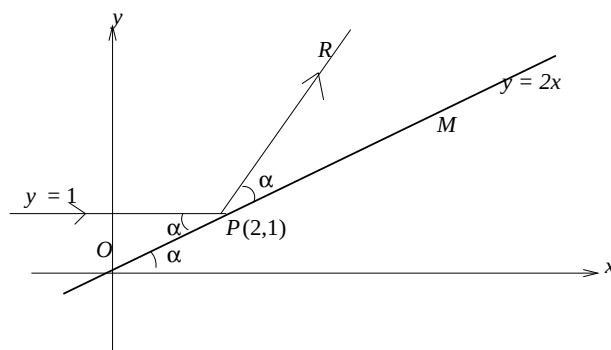
(A) $4x - 3y = 5$

(B) $3x - 4y = 2$

(C) $x - y = 1$

(D) $2x - 3y = 1$

Answer and Comments: (A). The best way to understand such problems is to draw a diagram such as the one below where the mirror is shown by a thick line M , the point of incidence of the light ray by P and the reflected ray by R .



Clearly, $P = (2, 1)$. So to find the equation of R we only need to find its slope, say m . This is done by the law of reflection which says that

the incident ray and the reflected ray make equal angles with the line of the mirror. This common angle is shown by α . (Normally, the law of reflection is stated to say that the incident and the reflected rays make equal angles with the normal. But that is needed when the mirror is a surface rather than a line. In the present case, making equal angles with the normal to M at P is equivalent to making equal angles with M itself.) Clearly,

$$\tan \alpha = \frac{1}{2} \quad (1)$$

Also the slope of M is $\frac{1}{2}$. Since R has slope m , we get

$$\tan \alpha = \frac{m - \frac{1}{2}}{1 + \frac{m}{2}} = \frac{2m - 1}{m + 2} = \frac{1}{2} \quad (2)$$

Solving, $m = \frac{4}{3}$ and the equation of R is,

$$y - 1 = \frac{4}{3}(x - 2) \quad (3)$$

which simplifies to $4x - 3y = 5$. So (A) is correct. ■

A very simple problem for those who know the law of reflection. the formula for the angle between two lines with slopes m_1 and m_2 is often stated as $\tan^{-1} \frac{|m_2 - m_1|}{1 + m_1 m_2}$. Those who apply it blindly will get two values of m because there are two angles between the two lines. They will then have to discard one. But if we keep in mind that we are measuring the acute angle α between M and R and M has a higher slope than R , then this ambiguity can be avoided by using (2).

In no other options given the line R has slope $\frac{4}{3}$. So a clever student may simply get the correct answer as soon as he gets the slope of R correctly. The paper setters should have been careful to include at least one fake option where the slope is $\frac{4}{3}$.

Q.7 For a real number x , let $[x]$ denote the greatest integer less than or equal to x . Then the number of real solutions of $|2x - [x]| = 4$ is

- (A) 1 (B) 2 (C) 3 (D) 4.

Answer and Comments: (D). Dropping the absolute value, the equation can be rewritten as

$$2x - [x] = \pm 4 \quad (1)$$

which splits into two separate equations. viz.

$$2x = [x] + 4 \quad (2)$$

$$\text{and } 2x = [x] - 4 \quad (3)$$

We now rewrite x as $[x] + \{x\}$ where $\{x\}$ is the fractional part of x . It always lies in the semi-open interval $[0, 1)$. This reduces (2) and (3) to

$$[x] + 2\{x\} = 4 \quad (4)$$

$$\text{and } [x] + 2\{x\} = -4 \quad (5)$$

respectively. In both the equations all terms other than $2\{x\}$ are integers. So $2\{x\}$ must also be an integer. As $\{x\} \in [0, 1)$, the only possibilities are $\{x\} = 0$ and $\{x\} = \frac{1}{2}$. We solve (4) and (5) separately with both the possibilities each.

In (4), if $\{x\} = 0$, then $[x] = 4$ and hence $x = 4$. But if $\{x\} = \frac{1}{2}$, then $[x] = 3$ and hence $x = \frac{7}{2}$.

Similarly, in (5), if $\{x\} = 0$, then $x = [x] = -4$. But if $\{x\} = \frac{1}{2}$, then $[x] = -5$ and so $x = -5 + \frac{1}{2} = -\frac{9}{2}$.

Summing up the given equation has four solutions. ■

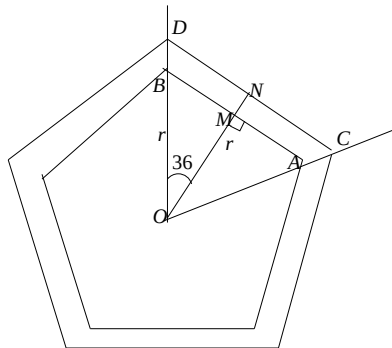
A simple and well designed problem requiring the use of the fractional part too.

Q.8 What is the ratio of the areas of the regular pentagons inscribed inside and circumscribed around a given circle?

$$(A) \cos 36^\circ \quad (B) \cos^2 36^\circ \quad (C) \cos^2 54^\circ \quad (D) \cos^2 72^\circ$$

Answer and Comments: (B). Let r and O be the radius and the centre of the circle. The inscribed and circumscribing pentagons are shown in the figure below so that their corresponding vertices are collinear with the centre. (This is not given to be the case in the statement

of the problem. But since the area of a regular pentagon inscribed in a circle is independent of where its vertices lie, we are free to choose them.)



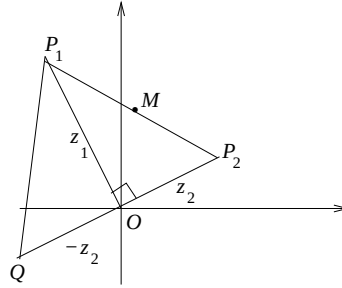
Take two adjacent vertices A, B of the inscribed pentagon and let C, D be the corresponding vertices of the circumscribing pentagon. Let the angle bisector of $\angle AOB$ meet AB and CD at M and N respectively. The ratio of the area of the inner pentagon to the outer one is the same as the ratio of the areas of the triangle OMB to OND . And since these triangles are similar, this ratio is the square of the ratio of the lengths of the corresponding sides OM and ON . We calculate these sides. Clearly $ON = r$. To find OM , we note that $\angle AOB = 72^\circ$ and hence $\angle MOB = 36^\circ$. In the right angled triangle OMB , $OB = r$ and so $OM = r \cos 36^\circ$. Therefore $OM : ON = \cos 36^\circ$. So the ratio of the areas of the two pentagons is $\cos^2 36^\circ$. ■

A simple trigonometric problem, based on the fact that the ratio of the areas of two similar figures is the square of the ratio of their corresponding sides. It is good that the answer has been asked in terms of $\cos 36^\circ$. Its exact numerical value is $\frac{\sqrt{5} + 1}{4}$, but is not as well known or frequently used as the values of, say $\cos 45^\circ$ or $\sin 30^\circ$. So those who remember it get no special advantage.

Q.9 Let z_1, z_2 be nonzero complex numbers satisfying $|z_1 + z_2| = |z_1 - z_2|$. The circumcentre of the triangle with the points z_1, z_2 and the origin as its vertices is given by

- (A) $\frac{1}{2}(z_1 - z_2)$ (B) $\frac{1}{3}(z_1 + z_2)$ (C) $\frac{1}{2}(z_1 + z_2)$ (D) $\frac{1}{3}(z_1 - z_2)$.

Answer and Comments: (C). In the Argand diagram let P_1, P_2 be the points representing the complex numbers z_1 and z_2 respectively. Let Q represent $-z_2$ as shown below.



Then $|z_1 - z_2|$ and $|z_1 + z_2|$ are the lengths of the segments P_1P_2 and P_1Q respectively and so the given condition means that P_1 lies on the perpendicular bisector of the segment QP_2 . But the midpoint of QP_2 is O , the origin. So, P_1OP_2 is a right angled triangle. Therefore its circumcentre is at the midpoint, say M , of P_1P_2 . It corresponds to the complex number $\frac{1}{2}(z_1 + z_2)$. ■

The wording of the problem makes it abundantly clear that it is more a problem in geometry than about complex numbers. The problem is simple once the given condition is interpreted geometrically. Then the calculations involved are simple. This is therefore a good problem as an MCQ.

Those who cannot think of the geometric approach immediately can begin by analysing the given condition, viz.

$$|z_1 + z_2| = |z_1 - z_2| \quad (1)$$

in a purely algebraic manner. Squaring both the sides

$$(z_1 + z_2)(\overline{z_1} + \overline{z_2}) = (z_1 - z_2)(\overline{z_1} - \overline{z_2}) \quad (2)$$

Expanding both the sides

$$|z_1|^2 + |z_2|^2 + z_1\overline{z_2} + z_2\overline{z_1} = |z_1|^2 + |z_2|^2 - z_1\overline{z_2} - z_2\overline{z_1} \quad (3)$$

which means

$$z_1 \overline{z_2} + z_2 \overline{z_1} = 0 \quad (4)$$

But that means

$$|z_1 - z_2|^2 = |z_1|^2 + |z_2|^2 \quad (5)$$

So the triangle with vertices at z_1, z_2 and the origin is right angled at O . To finish the solution, you still need the geometric result about the circumcentre of a right angled triangle. But now the need for it far more obvious.

- Q.10 In how many ways can 20 identical chocolates be distributed among 8 students so that each student gets at least one chocolate and exactly two students get at least two chocolates each?

(A) 308 (B) 364 (C) 616 (D) $\binom{8}{2} \binom{17}{7}$

Answer and Comments: (A). First give one chocolate to each student. We are left with 12 chocolates and now the restriction left is that exactly two students should get at least one chocolate. That is equivalent to saying that all 12 chocolates are to go to these two ‘privileged’ students but not all 12 should go to either one of them.

This simplified version leads to an easy solution. The two privileged students can be selected in $\binom{8}{2} = 28$ ways. Between these two students the distribution is $1 + 11$ or $2 + 10$ or $\dots$ $10 + 2$ or $11 + 1$. So there are 11 possible distributions. Hence the total number of distributions is $28 \times 11 = 308$. ■

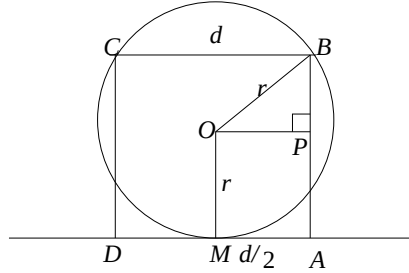
The key idea is to reduce the problem to a simpler one by first giving one chocolate to each student. However, even for those who get this idea, the paper setters have been clever enough to give (D) as a tempting option. That makes the problem simple but well designed. If instead of ‘at least two’, the phrase ‘more than one’ were used it would have made it a little easier to grasp the key idea. Because then one can think of giving one chocolate to each student, dismiss all but two

students, and distribute the remaining 12 chocolates between the two who are kept. Interesting how a slight change of diction can make a problem a little easier.

- Q.11 Two vertices of a square lie on a circle of radius r , and the other two vertices lie on a tangent to this circle. Then, each side of the square is

(A) $\frac{3r}{2}$ (B) $\frac{4r}{3}$ (C) $\frac{6r}{5}$ (D) $\frac{8r}{5}$.

Answer and Comments: (D). Another simple geometry problem the right start for whose is by drawing a diagram. It is shown below.



Let O be the centre of the circle, $ABCD$ the square with A and D lying on the tangent to the circle and B, C lying on the circle. Let M be the midpoint of AD . Then M is also on the circle and so

$$OM = r \quad (1)$$

Drop perpendicular OP to the side AB . Then OP is parallel to AD and so

$$PA = r \quad (2)$$

Now let d be the side of the square $ABCD$. Then

$$BP = BA - PA = d - r \quad (3)$$

Also

$$OP = MA = \frac{d}{2} \quad (4)$$

So, from the right angled triangle OPB we have

$$r^2 = \left(\frac{d}{2}\right)^2 + (d - r)^2 \quad (5)$$

which simplifies to

$$\frac{5}{4}d^2 = 2rd \quad (6)$$

As d is obviously non-zero, we get $d = \frac{8r}{5}$. ■

A good problem to be asked as an MCQ. But it would be a more challenging problem to construct, given a circle of radius r , such a square using straight edge and compass only. In that case, we first use the analysis above to get hold of the side d as $\frac{8r}{5}$. Hence MA equals half the side, i.e. $\frac{4r}{5}$. So the construction would begin by taking any point M on the circle, drawing through M a line perpendicular to OM and then marking two points A, D on this line which are at a distance $\frac{4r}{5}$ each from M . The remaining two vertices can now be found by drawing lines through A and D which are perpendicular to AD and marking points B, C on them at distances $\frac{8r}{5}$ from A, D . Among all problems of geometry, those asking for straight edge and compass constructions require considerable imagination, based on some analysis first. The present problem involves only the analysis. The MCQ format automatically precludes construction problems. But they are ideal for Paper 2.

- Q.12 Let P be the set of all numbers obtained by multiplying five distinct integers between 1 and 100. What is the largest integer n such that 2^n divides at least one element of P ?

(A) 8 (B) 20 (C) 24 (D) 25

Answer and Comments: (C). This is one of those problems which are mathematically simple but appear difficult because of their clumsy wording. We are dealing with numbers of the form $x = x_1x_2x_3x_4x_5$

where $x_1, x_2, \dots, x_5$ are distinct integers between 1 and 100. For each such integer x , we have to find the highest power of 2 which divides it and then we have to maximise this highest power by varying x over P .

Obviously, the highest power of 2 which divides x has exponent $k_1 + k_2 + k_3 + k_4 + k_5$ where 2^{k_i} is the highest power of 2 which divides x_i for $i = 1, 2, 3, 4, 5$. Note that even though $x_i \neq x_j$ for $i \neq j$, it may happen that $k_i = k_j$. This happens when x_i and x_j are odd multiples of 2^{k_i} . For example, the highest power of 2 which divides 40 is the same as the highest power of 2 which divides 24. But the highest power of 2 which divides 48 is greater because 48 is an even multiple of 2.

Once this point is understood, it is easy to arrive at the answer. In each x_i we try to accommodate the highest power of 2 as a factor, subject to the given restrictions. The highest power of 2 not exceeding 100 is $64 = 2^6$. The only odd multiple of it lying between 1 and 100 is 64 itself. The next in rank is $32 = 2^5$. It has two odd multiples in the given range, viz. 32 and 96. We already have three numbers. Let us look for the remaining two from $16 = 2^4$. This time there are three possible candidates, viz. 16, 48 and 80. But we have room only for two of them. It does not matter which two we select because the only thing that matters is the highest power of 2 that divides them. For simplicity we take these integers as 16 and 48. So, now we have got five integers, viz. 64, 32, 96, 16 and 48 whose product will be divisible by 2^n for the highest possible value of n . By a direct calculation, $n = 6 + 5 + 5 + 4 + 4 = 24$. ■

The problem is more a test of the ability to understand a given problem. This is a commendable feature. Once in a while there should be problems of this type, unlike the more common ones where it takes no thinking to understand the problem (e.g. a problem about finding solutions of some equation).

Those who miss that distinct odd multiples of the same power of 2 are allowed, will take the five numbers as 64, 32, 16, 8 and 4. The answer they will get then is $6 + 5 + 4 + 3 + 2 = 20$. The paper setters have apparently thought of this possibility because they have indeed given 20 as an option.

Q.13 Consider the function $f(x) = ax^3 + bx^2 + cx + d$, where a, b, c and d are

real numbers with $a > 0$. If f is strictly increasing, then the function $g(x) = f'(x) - f''(x) + f'''(x)$ is

- (A) zero for some $x \in \mathbb{R}$ (B) positive for all $x \in \mathbb{R}$
 (C) negative for all $x \in \mathbb{R}$ (D) strictly increasing.

Answer and Comments: (B). By a direct calculation,

$$\begin{aligned} g(x) &= f'(x) - f''(x) + f'''(x) \\ &= (3ax^2 + 2bx + c) - (6ax + 2b) + 6a \\ &= 3ax^2 + (2b - 6a)x + (c - 2b + 6a) \end{aligned} \quad (1)$$

Since $g(x)$ is a quadratic with positive leading coefficient, options (C) and (D) are ruled out. To see which of the remaining two options holds, we need to check if $g(x)$ has any real roots. For this we consider its discriminant, say Δ which equals

$$\Delta = 4[(b - 3a)^2 - 3a(c - 2b + 6a)] = 4(b^2 - 3ac - 9a^2) \quad (2)$$

We are given that f is strictly increasing. So its derivative $f'(x)$ has no real roots. Hence its discriminant cannot be positive. That means

$$4b^2 \leq 12ac \quad (3)$$

or $b^2 < 3ac$. But then $\Delta < 0$ by (2). So (A) is false and only (B) holds.

■

A straightforward problem, based more on the roots of a quadratic. There was no need to give that $a > 0$. Without this, $f(x)$ will not be strictly increasing.

Q.14 Let A be the set of all points (h, k) such that the area of the triangle formed by (h, k) , $(5, 6)$ and $(3, 2)$ is 12 square units. What is the least possible length of a line segment joining $(0, 0)$ to a point in A ?

- (A) $\frac{4}{\sqrt{5}}$ (B) $\frac{8}{\sqrt{5}}$ (C) $\frac{12}{\sqrt{5}}$ (D) $\frac{16}{\sqrt{5}}$

Answer and Comments: (B). We first need to identify the set A . From the formula for the area of a triangle with given vertices, the given condition means

$$\begin{vmatrix} h & k & 1 \\ 5 & 6 & 1 \\ 3 & 2 & 1 \end{vmatrix} = \pm 24 \quad (1)$$

Expanding the determinant, this resolves into two equations

$$4h - 2k = 32 \quad (2)$$

$$\text{and } 4h - 2k = -16 \quad (3)$$

Each equation represents a straight line and A is their union. The perpendicular distance from O to them are $\frac{32}{\sqrt{20}} = \frac{16}{\sqrt{5}}$ and $\frac{16}{\sqrt{20}} = \frac{8}{\sqrt{5}}$. Clearly the second one is the smaller. So the closest point of A to the origin is at a distance $\frac{8}{\sqrt{5}}$ from it. ■

Another absolutely straightforward problem.

Q.15 Let $P = \{abc : a, b, c \text{ positive integers, } a^2 + b^2 = c^2, \text{ and } 3 \text{ divides } c\}$. What is the largest integer n such that 3^n divides every element of P ?

(A) 1 (B) 2 (C) 3 (D) 4

Answer and Comments: (D). This problem is of a similar spirit as Q.12, except that now we are dealing with a product of three positive integers a, b, c instead of five and the restrictions on these three integers are of a very different type. In Q.12, the restriction was merely that all five factors were from the set $\{1, 2, \dots, n\}$. This time the restriction is that c is divisible by 3. But there is no restriction as to how big a, b, c can be. Also we have to find the largest power of 3 which divides abc in *every* such triple (a, b, c) . Obviously, since 3 divides c , it divides abc too and so the answer is at least 1. But can it be higher?

To answer this we first prove that if

$$a^2 + b^2 = c^2 \quad (1)$$

and 3 divides c then 3 must divide both a and b . For otherwise a and b will be of the form $a = 3k \pm 1$ and $b = 3r \pm 1$ for some integers k, r . Squaring and adding, we get

$$c^2 = a^2 + b^2 = 9(k^2 + r^2) \pm 6k \pm 6r + 2 \quad (2)$$

All the terms on the R.H.S. are divisible by 3 except 2. But that would mean c^2 is not divisible by 3, and hence that 3 divides c (as 3 is a prime), a contradiction.

Thus we have shown that under the hypothesis of the problem, divisibility of c by 3 implies that of both a and b . As a result the product abc is divisible by 27. So the exponent n for which 3^n divides abc is at least 3.

But we can go one step further. Let $a' = a/3, b' = b/3$ and $c' = c/3$. All of them are integers and they satisfy

$$a'^2 + b'^2 = c'^2 \quad (3)$$

3 may or may not divide c' . If it does then $abc = 3^3 a' b' c'$ is divisible by 3^4 . But we shall show that if (3) holds, then regardless of whether 3 divides c' , at least one of a' and b' is divisible by 3. This will give us one extra power of 3 in abc and raise the exponent n from 3 to 4. To see this, note first that every integer m is of the form $3u$ or $3u \pm 1$ for some integer u . Then $m^2 = 9u^2$ or $m^2 = 9u^2 \pm 6u + 1$ which is of the form $3k + 1$. Applying this to (1), if 3 divides neither a' nor b' , then $a'^2 = 3r + 1$ and $b'^2 = 3s + 1$ for some integers r, s . But then $a'^2 + b'^2 = 3(r + s) + 2$ which cannot be c'^2 . So the highest exponent n such that 3^n divides abc is at least 4.

Technically, we must also show that we cannot do better than 3^4 . For this consider a triplet (12, 9, 15). Here the product abc is divisible only by 3^4 and not by 3^5 . ■

A good problem which is more suitable for Paper 2 where justifications are needed. The essential idea is that a square can be of the form $3k$ or $3k + 1$ but never of the form $3k + 2$. This result is of the same spirit as a more well known result that a square can never be of the form $4k + 2$ or $4k + 3$.

Although not needed for this solution, it is instructive to have a deeper analysis of (1). For obvious reasons, triples (a, b, c) of positive integers satisfying (1) are called **Pythagorean triples**. If k is a common factor of any two of out of a, b, c , it will divide the third one too and then we would get $(\frac{a}{k}, \frac{b}{k}, \frac{c}{k})$ as another Pythagorean triple. So weeding out common factors, if any, we are left with a Pythagorean triple (a, b, c) in which every two members are coprime. (In analogy with reduced rationals, such Pythagorean triples are called **reduced Pythagorean triples**.) It is not hard to show that in a reduced triple (a, b, c) , exactly one of a and b is even. Without loss of generality we suppose a is even. Then it can be shown that there exist positive integers u, v such that

$$a = 2uv, b = u^2 - v^2 \text{ and } c = u^2 + v^2 \quad (4)$$

(For a proof see Exercise (4.22). The familiar $(4, 3, 5)$ triple arises from $u = 2, v = 1$ while $(12, 7, 13)$ comes from $u = 4, v = 3$. $u = 4, v = 1$ gives a relatively less common triple $(8, 15, 17)$.)

Q.16 Let $A_0 = \emptyset$ (the empty set). For each $i = 1, 2, 3, \dots$, define the set $A_i = A_{i-1} \cup \{A_{i-1}\}$. Then the set A_3 is

- (A) $\emptyset$ (B) $\{\emptyset\}$ (C) $\{\emptyset, \{\emptyset\}\}$ (D) $\{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$

Answer and Comments: (D). There is a very simple way to see this. A_0 has no elements in it. For every $i > 0$, A_i is obtained by adding one more element, viz. A_{i-1} to the set A_{i-1} . Note that this new element is not present in the set A_{i-1} . So, every time we go from A_{i-1} to A_i , the number of elements goes up by 1. Therefore A_i has i elements for every $i \geq 0$. In particular, A_3 has 3 elements. So, (D) is correct. ■

The construction of these sets is historically important in mathematics. There was a view held for a long time that God gave us natural numbers and man constructed everything else such the integers, the rational numbers, the real numbers and the complex numbers, step by step. But if we define the sets A_i as above, then they can be taken as the very construction of natural numbers.

Q.17 Let $f(x) = \frac{1}{x-2}$. The graphs of the functions f and f^{-1} intersect at

- (A) $(1 + \sqrt{2}, 1 + \sqrt{2})$ and $(1 - \sqrt{2}, 1 - \sqrt{2})$
 (B) $(1 + \sqrt{2}, 1 + \sqrt{2})$ and $(\sqrt{2}, -1 - \frac{1}{\sqrt{2}})$
 (C) $(1 - \sqrt{2}, 1 - \sqrt{2})$ and $(-\sqrt{2}, -1 + \frac{1}{\sqrt{2}})$
 (D) $(\sqrt{2}, -1 - \sqrt{2})$ and $(\sqrt{2}, -1 + \frac{1}{\sqrt{2}})$

Answer and Comments: (A). If a function $y = f(x)$ is one to and onto, then the inverse function f^{-1} exists. It is given by $x = f^{-1}(y)$. So generally, x is expressed as a function of y . But in this problem, we have to express both f and f^{-1} as functions of x . As a very common example, if $f(x) = \sin x$ where $x \in [-\pi/2, \pi/2]$ then its inverse function is $f^{-1}(x) = \sin^{-1} x$ where $x \in [-1, 1]$. As other examples, if $f(x) = x^3$ then $f^{-1}(x) = x^{1/3}$.

Coming to the problem, we are given

$$xy - 2y = 1 \tag{1}$$

which gives

$$x = \frac{1 + 2y}{y} = 2 + \frac{1}{y} = f^{-1}(y) \tag{2}$$

So replacing y by x ,

$$f^{-1}(x) = 2 + \frac{1}{x} \tag{3}$$

At a point of intersection, we must have $f(x) = f^{-1}(x)$. This gives an equation $\frac{1}{x-2} = 2 + \frac{1}{x}$ which simplifies to a quadratic in x , viz.

$$x^2 - 2x - 1 = 0 \tag{4}$$

whose roots are $1 \pm \sqrt{2}$. So (A) is correct. ■

A simple problem, once you realise that both f and f^{-1} are to be expressed as functions of x . The graphs of f and f^{-1} are symmetric w.r.t. the line $y = x$. So it is tempting to think that they meet only on that line. Luckily, that approach works for the present problem. But

in general, the two graphs can intersect at many other points too. If (x_0, y_0) is a point of intersection then so is (y_0, x_0) . But that does not mean $x_0 = y_0$. For example, if $f(x) = 1 - x^3$, then $f^{-1}(x) = (1 - x)^{1/3}$. Their graphs intersect at 5 points of which only one lies on the line $y = x$.

- Q.18 Let N be a number such that whenever you take N consecutive positive integers, at least one of them is coprime to 374. What is the smallest possible value of N ?

(A) 4 (B) 5 (C) 6 (D) 7

Answer and Comments: (C). A set of consecutive integers is often called a ‘block’. Although this word is not used in the question, we shall use it because of its graphic connotation. Coprime means not having any common factors (other than 1, of course). In the present problem 374 factors as $2 \times 11 \times 17$. So we want the block length N to be such that for every positive integer n , the integers from n to $n + N - 1$ contain at least some number which is not divisible by any of 2, 11 and 17. Consider any block of N consecutive positive integers. Nearly $N/2$ elements of it will be even and hence fail to be coprime to 374. Out of the odd elements in the block, if each one of them is divisible by either 11 or 17, then no member in the block will be coprime to 374. Of course, any two distinct odd multiples of 11 differ by at least 22 and so in any two consecutive odd integers, at least one will be coprime to 11. The same is true as far as being coprime to 17 is concerned. But it may happen that out of two consecutive odd numbers, one is a multiple of 11 and the other a multiple of 17. In that case, both will be disqualified, along with their three even neighbours and thus we shall get a block of five consecutive integers which contains no integer coprime to 374.

So we hunt for two such consecutive odd numbers. By trial, 119 is a multiple of 17 while 121 is a multiple of 11. So the block of five integers from 118 to 122 fails to have any integer coprime to 374. Hence the answer to the problem is at least 6. But it is easy to see that every block of six consecutive integers contains at least one number coprime to 374. This is so because such a block contains three odd integers and at most two will be multiples of 11 or 17. So (C) is correct. ■

As in the case of Q.10, a slight change of framing would have made the key idea more easily accessible. Instead of asking to find the smallest block size with a certain property, it would have been better to ask to find a longest block of consecutive integers in which every integer has at least one common factor with 374. Once you identify such a block, the N asked in the problem is obtained simply by adding 1 to the length of this block.

A very imaginative problem based on elementary facts about coprimality. The solution requires the factorisation of 374. But that is easy because by the rule of 11 (viz. that the sums of alternate digits should differ by a multiple of 11), we see that 374 is divisible by 11 since $3 + 4 = 7$. It does take some trial and error to come up with the numbers 119 and 121. But considering that 119 appears in the multiplication table of 17 while 121 is familiar as 11^2 , the search is not unreasonably chancy. We merely go on taking odd multiples of 17 and see if either of their immediate odd neighbours (e.g. 49 and 53 or 83 and 87) is a multiple of 11. Although the reasoning given here takes a long time to write down, it does not take much time to conceive the essential ideas in it. So this problem is well suited as an MCQ. In a conventional examination, justification can be asked. In the present problem the justification is mostly verbal rather than based on formulas and equations. So such a problem, if asked in Paper 2 will be a good test of the ability to articulate one's thoughts, which has declined considerably with exclusive reliance on MCQ's.

Actually, the odd multiples 119 and 121 of 17 and 11 respectively could also have been arrived at by a systematic procedure without resorting to trial. In essence we want two integers m and n such that

$$(2m + 1) \times 17 - (2n + 1) \times 11 = \pm 2 \quad (5)$$

which is equivalent to

$$m \times 17 - n \times 11 = -2 \text{ or } -4 \quad (6)$$

We begin by expressing 1 as a linear combination of 17 and 11 with integer coefficients. The simplest such combination that comes to mind is

$$2 \times 17 - 3 \times 11 = 1 \quad (7)$$

Multiplying throughout by -2 we get

$$(-4) \times 17 - (-6) \times 11 = -2 \quad (8)$$

Comparing this with the first possibility in (7), we can take $m = -4$ and $n = -6$. So $2m + 1 = -7$ and $2n + 1 = -11$. With these values, (6) becomes

$$-7 \times 17 - (-11) \times 11 = 2 \quad (9)$$

(Note that we are taking only the first possibility on the R.H.S. of (7).) Written differently,

$$11 \times 11 - 7 \times 17 = 2 \quad (10)$$

which is what we obtained earlier by trial. (Had we taken the second possibility on the R.H.S. of (7), viz. -4 , then we would have to multiply (8) throughout by -4 . The resulting m and n come out to be -8 and -12 respectively and the equation we get finally would be

$$15 \times 17 - 23 \times 11 = 2 \quad (11)$$

which goes to show that the two consecutive odd multiples that we are looking for are not unique. But that is no problem. We want only one. And now we can find it systematically from (8) instead of by mere trial.)

An immediate objection we have to face is that we are passing the buck, because (8) was obtained by trial too. So all we are doing is to avoid one trial by resorting to another! But there are two ways to answer this. First, (8) is a little easier to guess by trial than (11). More importantly, given two integers x and y whose greatest common divisor is d , there is systematic procedure to find integers a and b such that

$$ax + by = d \quad (12)$$

The g.c.d. of two integers x and y can be read out if we know their prime power factorisations. But this is a very difficult and time consuming task. (There is, in fact, no known efficient way to factorise a given integer into even two factors. All the secret passwords are designed on the assumption that nobody knows such an efficient method. If

this assumption fails, there will be a chaos as the passwords would be cracked very easily!) In any case, even if we are able to find d this way, it still remains a problem how to express in the form (12).

But in schools we learn that there is a simpler way to find the g.c.d. of two integers based on repeated applications of what is called **Euclid's division algorithm**. We begin by dividing the larger of the two numbers by the smaller one to get some remainder (which is necessarily smaller than the smaller of the two given numbers). We then divide the smaller number by this remainder to get a still smaller remainder. We repeat this process till we get 0 as the remainder, which is bound to happen in a finite number of steps since we are getting a strictly decreasing sequence of positive integers. (That is why the procedure is called an **algorithm**. Guaranteed termination in a finite number of steps is a *sine qua non* of any algorithm.) The last divisor is the g.c.d. of the original two numbers.

Familiar as it is, we illustrate this procedure by an example. Suppose we want to find the g.c.d. of 1092 and 195. We proceed as follows.

$$\begin{aligned} 1092 &= 5 \times 195 + 117 \\ 195 &= 1 \times 117 + 78 \\ 117 &= 1 \times 78 + 39 \\ 78 &= 2 \times 39 + 0 \end{aligned} \tag{13}$$

So the g.c.d. of 1092 and 195 is 39. What is generally not done in schools is to use the equations above in a backward manner (starting with the last but one equation) so as to express the g.c.d. as a linear combination of the original numbers. In the illustration above. This can be done as follows.

$$\begin{aligned} 39 &= 117 - 1 \times 78 \\ &= 117 - 1 \times (195 - 1 \times 117) = 2 \times 117 - 1 \times 195 \\ &= 2 \times (1092 - 5 \times 195) - 1 \times 195 \\ &= 2 \times 1092 + (-11) \times 195 \end{aligned} \tag{14}$$

So we have expressed the g.c.d. of 1092 and 195 as a linear combination of them. If we apply this procedure to the pair 11 and 17, we have

$$17 = 1 \times 11 + 6$$

$$\begin{aligned}
11 &= 1 \times 6 + 5 \\
6 &= 1 \times 5 + 1 \\
5 &= 5 \times 1 + 0
\end{aligned} \tag{15}$$

So, the g.c.d. of 17 and 11 is 1, which we know already since they are distinct primes. But now we can express 1 as a linear combination of 17 and 11 by reversing the steps. As in the earlier illustration we start with the last but one equation. Thus,

$$\begin{aligned}
1 &= 6 - 1 \times 5 \\
&= 6 - 1 \times (11 - 1 \times 6) = 2 \times 6 - 1 \times 11 \\
&= 2 \times (17 - 1 \times 11) - 1 \times 11 \\
&= 2 \times 17 - 3 \times 11
\end{aligned} \tag{16}$$

which is exactly the same as (8) which we got earlier by trial. That may not always happen because in general there are many different ways to express the g.c.d as a linear combination of the two numbers. For example, the g.c.d. 1 of 5 and 3 can be written either as $2 \times 3 - 1 \times 5$ or as $2 \times 5 - 3 \times 3$. The stress is not on uniqueness but on existence of a linear combination.

As a result, the question can be asked as a full length question if we replace 374 by $2pq$ where p, q are any two distinct (odd) primes. In the proof the candidate does not have to actually find two consecutive odd numbers one of which is divisible by p and the other by q . Proving their existence from the coprimality of p and q in the manner above and thereby arriving at the same minimum block length (viz. 6) would be an interesting question.

- Q.19 Let $A_1, A_2, \dots, A_{18}$ be the vertices of a regular polygon with 18 sides. How many of the triangles $\triangle A_i A_j A_k$, $1 \leq i < j < k \leq 18$, are isosceles but not equilateral?

(A) 63 (B) 70 (C) 126 (D) 144

Answer and Comments: (C). In any triangle which is isosceles but not equilateral, there is a unique ‘middle’ vertex where the equal sides meet. This middle vertex could be any one of the 18 vertices of the 18-gon. Fix it to be A_1 . Then one of the remaining two vertices of

the triangle will lie somewhere from A_2 to A_9 except at A_7 (since the triangle is not equilateral). Once this vertex is fixed, the third vertex is uniquely determined. So the total number of desired triangles is $18 \times 7 = 126$. ■

Another simple counting problem. It is good that the paper setters have specifically excluded equilateral triangles. Otherwise a controversy can arise because in that case the correct answer will not be $126 + 18$ but $126 + 6$ because each equilateral triangle gets counted thrice in the method above. In fact that would have been a more interesting problem.

Q.20 The limit $\lim_{x \rightarrow 0} \frac{\sin^\alpha x}{x}$ exists *only when*

- (A) $\alpha \geq 1$ (B) $\alpha = 1$
 (C) $|\alpha| \leq 1$ (D) α is a positive integer.

Answer and Comments: (A). For $\alpha = 1$, the limit in the question is $\lim_{x \rightarrow 0} \frac{\sin x}{x}$. It exists and its value 1. It is the basic limit needed to find the derivatives of the trigonometric functions. Surely, if $\alpha > 1$, we can write $\alpha = 1 + h$ where $h > 0$. Then $\frac{\sin^\alpha x}{x}$ will equal $\frac{\sin x}{x} (\sin x)^h$ and this product will tend to 0 because of the second factor $(\sin x)^h$. The question uses the words ‘only when’ emphatically. So even though the limit exists in (B) and (D) too, they put an unnecessarily stronger restriction on α and hence are eliminated. (C) is eliminated because the limit in question does not exist for $\alpha < 1$. ■

A very simple problem based on the existence of the limit of $\frac{\sin x}{x}$ as x tends to 0.

Q.21 Consider the region $R = \{(x, y) : x^2 + y^2 \leq 100, \sin(x + y) > 0\}$. What is the area of R ?

- (A) 25π (B) 50π (C) 50 (D) $100\pi - 50$

Answer and Comments: (B). The given region R is a subset of a disc, say D , of radius 10 centred at the origin. Whether a point $(x, y) \in D$ lies in R not depends only on $x + y$. That means if some point (x_1, y_1) is in R then so will be all points (x, y) in D for which $x + y = x_1 + y_1$. So R will consist of the union of segments of the intersections with D of the lines of the form $\{(x, y) : x + y = c\}$ where c is some constant. Let us first see the range of c is as (x, y) varies over D . Obviously $|x + y|$ will increase as we move away from the centre O . So the maximum of $x + y$ will occur at some point (x_0, y_0) in the first quadrant lying on the boundary circle $S = \{(x, y) : x^2 + y^2 = 100\}$ and the minimum will occur at its antipodal point $(-x_0, -y_0)$ in the third quadrant. To locate these points, parametrise the first quadrant portion of S by

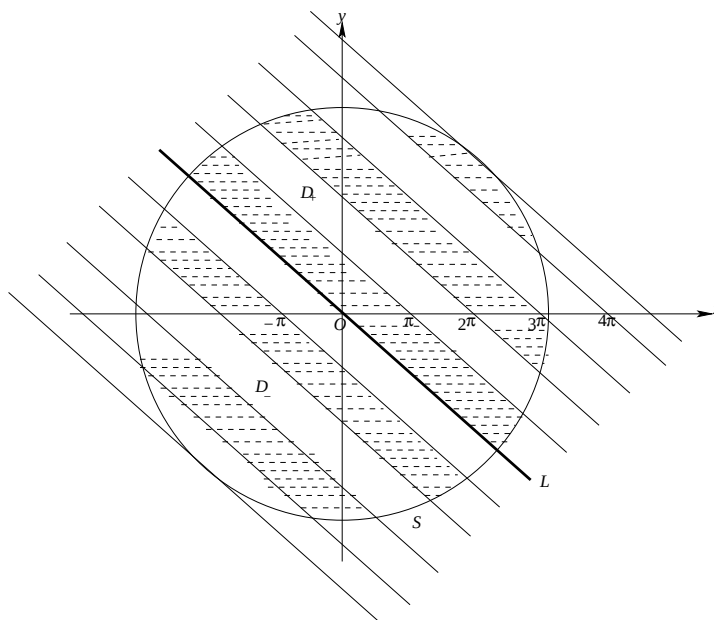
$$x = 10 \cos \theta, y = 10 \sin \theta, 0 \leq \theta \leq \frac{\pi}{2} \quad (1)$$

Then

$$x + y = 10(\cos \theta + \sin \theta) = 10\sqrt{2} \cos(\theta - \frac{\pi}{4}) \quad (2)$$

whose maximum occurs when $\theta = \frac{\pi}{4}$. So as (x, y) varies over D , $x + y \in [-10\sqrt{2}, 10\sqrt{2}]$. (The lovers of calculus can, of course, maximise $\cos \theta + \sin \theta$ by taking its derivative too. They can hardly be blamed for missing this short cut. Love is blind, after all!)

Now, $\sin(x + y) > 0$ if and only if $2k\pi < x + y < (2k + 1)\pi$ for some integer k . So the region R splits into several parallel strips (shown by shading in the figure below). Consider the line $L = \{(x, y) : x + y = 0\}$. This line, shown thickly, is very important for us. It is a diameter of D and hence divides it into two equal parts. Then the portions of R lying on one side of R are $\{(x, y) \in D : 0 < x + y < \pi\}$, $\{(x, y) \in D : 2\pi < x + y < 3\pi\}$ and finally, $\{(x, y) \in D : 4\pi < x + y \leq 10\sqrt{2}\}$. (We are using here that $4\pi < 10\sqrt{2} < 5\pi$ which can be justified by taking approximate values of π and $\sqrt{2}$.) On the other side of the line L , R contains points of the strips, $\{(x, y) : -2\pi < x + y < -\pi\}$ and $\{(x, y) \in D : -4\pi < x + y < -3\pi\}$. Note that the boundaries of all these strips are line segments parallel to L (along with some arcs of the circle $S = \{(x, y) : x^2 + y^2 = 100\}$.)



It will be a horrendous task to calculate the area of each of these five strips and add them. But there is a way out. Since the sine function is odd, if $(x, y) \in R$, then $(-x, -y) \notin R$. Geometrically, for every strip on one side of L which is included in R , there is a congruent strip on the other side of L which is in the complement of R . So R and its complement in the disc D have the same areas. Together their areas add to the area of the disc, which is 100π . So the area of R (and also of its complement in D) is 50π . ■

Although the problem asks for the area of a region, nowhere are integrals involved in its calculation. Instead, the success lies in an imaginative use of the fact that the line L divides D into two parts and the fact that the sine function is an odd function. In principle, this is analogous to the evaluation of the definite integral I in Q.5, where also we saw that although I is hard to calculate directly, if we take a suitable copy J of it, then $I + J$ is easy to evaluate. From the options 25π and 50π a candidate is likely to get the hint that the area of R is some fraction of the area of D and such a candidate will be rewarded if he tries to go into the reason. So this is a very well designed question which leads the good candidates to the key idea.

In fact, once the key idea (based on the line L and the oddness of the sine function) strikes, it is hardly necessary to split the region R into five strips as we have done. Instead one can argue as follows. The line L splits the disc into two parts D_+ and D_- where $D_+ = \{(x, y) \in D : x + y \geq 0\}$ and $D_- = \{(x, y) \in D : x + y \leq 0\}$. We let $R_+ = R \cap D_+$ and $R_- = R \cap D_-$. Then $R_+ \cap R_-$ has zero area because the intersection is contained in the line L which has zero area. (Even this is not needed because by stipulating that $\sin(x + y) > 0$, the paper setters have carefully excluded all points on L .) So, if we use A to denote the area, then we have

$$A(R) = A(R_+) + A(R_-) \quad (3)$$

Now let R' be the complement of R in D , i.e. $R' = \{(x, y) \in D : (x, y) \notin R\}$. Then we have a similar decomposition of R' into R'_+ and R'_- and get

$$A(R') = A(R'_+) + A(R'_-) \quad (4)$$

Now let $f(x, y) = (-x, -y)$. Then f maps D onto itself. The function f , being the reflection into the origin is a congruence and hence preserves areas. Moreover, if $(x, y) \in R$ then $f(x, y) = (-x, -y) \in R'$ because $\sin(-x - y) = -\sin(x + y)$. So f maps R congruently onto R' . (Strictly speaking, this is not true. Because of the strict inequality $\sin(x + y) > 0$, R does not contain the portions of the lines that bound these strips. So R' contains them. But each such boundary segment has zero area and so we might as well ignore them as far as areas are concerned. Anyway, in an MCQ, nobody bothers about such fussy points.)

Further if $(x, y) \in D_+$ then $f(x, y) \in D_-$, So f maps R_+ congruently onto R'_- and R_- congruently onto R'_+ . So we have

$$A(R_+) = A(R'_-) \quad (5)$$

$$\text{and } A(R_-) = A(R'_+) \quad (6)$$

Substituting these into (3) and using (4) we get

$$A(R) = A(R') \quad (7)$$

But D is the union of R and R' and $R \cap R' = \emptyset$, So

$$A(D) = A(R) + A(R') = 2A(R) \quad (8)$$

Since $A(D) = 100\pi$ we get $A(R) = 50\pi$. ■

Thus in a way the work done earlier of finding the range in which $x + y$ varies (viz. $[-10\sqrt{2}, 10\sqrt{2}]$) and then splitting R into five parallel strips was totally unnecessary. Still it has its value. It is very rare that the key idea flashes right at the start. Usually, it comes after putting in some but not necessarily the entire hard work. For example, after drawing the strip of R which is adjacent to the diametrical line L in the figure above, one may realise that its the adjacent strip on the other side of L is not in R and has the same area as the strip that is in R . In that case, it is hardly necessary to draw the entire figure or even to calculate the range of $x + y$ as (x, y) varies over D . Then when one goes into the reason for this it comes to mind that it lies in the fact that the reflection is a congruence and the sine function is odd.

The reason we have nevertheless completed the diagram of R is that if you see an elegant solution right at the start, you might not appreciate its beauty. You will do so only if you have first attempted the problem the hard way.

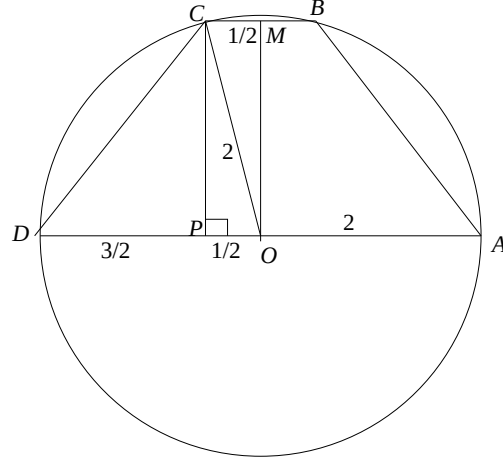
This question is better suited as an MCQ, because if a rigorous reasoning has to be given, one has to take into account the fact that the boundaries of the strips are in R' but not in R , So although f maps R congruently into R' , the mapping is not onto. One then has to argue that the extra portion of R' (viz. $R' - f(R)$) has zero area. Such fine points are incidental to the main theme.)

- Q.22 Consider a cyclic trapezium whose circumcentre is on one of the sides. If the ratio of the two parallel sides is $1 : 4$, what is the ratio of the sum of the two oblique sides to the longer parallel side?

(A) $\sqrt{3} : \sqrt{2}$ (B) $3 : 2$ (C) $\sqrt{2} : 1$ (D) $\sqrt{5} : \sqrt{3}$

Answer and Comments: (A). The side which contains the circumcentre must be a diameter of the circle and hence the longest side too. Let the trapezium be $ABCD$ with AD a diameter of the circle and O

the centre of the circle. Let M be the midpoint of the side BC . Drop a perpendicular CP to AB . Join OM, OC .



We take the lengths of AB and CD to be 4 units and 1 unit respectively. Then

$$CM = 1/2 \quad (1)$$

So,

$$CP = OM = \sqrt{OC^2 - CM^2} = \sqrt{4 - \frac{1}{4}} = \frac{\sqrt{15}}{2} \quad (2)$$

So, from the right angled triangle CPD ,

$$CD = \sqrt{CP^2 + DP^2} = \sqrt{\frac{15}{4} + \frac{9}{4}} = \sqrt{6} \quad (3)$$

Since $AB = CD$,

$$(AB + CD) : AD = 2\sqrt{6} : 4 = \sqrt{6} : 2 = \sqrt{3} : \sqrt{2} \quad (4)$$

So (A) holds. ■

An absolutely simple geometry problem requiring little more beyond Pythagoras theorem. As in the case of Q.11, a more interesting problem, suitable for Paper 2 would have been to construct a trapezium satisfying the conditions in the data.

Q.23 Consider the function $f(x) = \left\{ \log_e \left(\frac{4+\sqrt{2x}}{x} \right) \right\}^2$ for $x > 0$. Then,

- (A) f decreases upto some point and increases after that
- (B) f increases upto some point and decreases after that
- (C) f increases initially, then decreases and then again increases
- (D) f decreases initially, then increases and then again decreases.

Answer and Comments: (A). All the options depend on the sign of $f'(x)$. So we find $f'(x)$. We simplify the expression for $f(x)$ using the properties of logarithms. That way we shall be spared the differentiation of a quotient. Thus

$$f(x) = [\log_e(4 + \sqrt{2x}) - \log_e x]^2 \quad (1)$$

So,

$$f'(x) = 2(\log_e(4 + \sqrt{2x}) - \log_e x) \left(\frac{\frac{\sqrt{2}}{2\sqrt{2x}}}{(4 + \sqrt{2x}) \times 2\sqrt{2x}} - \frac{1}{x} \right) \quad (2)$$

To study the increasing/decreasing behaviour of $f(x)$ we first need to identify the zeros of $f'(x)$. Ignoring the constant factor, we check where each of the remaining factors of $f'(x)$ vanish.

Setting the first factor equal to 0 gives the equation

$$\log_e(4 + \sqrt{2x}) = \log_e x \quad (3)$$

which is equivalent to

$$4 + \sqrt{2x} = x \quad (4)$$

Taking the radical term on one side and squaring,

$$2x = (x - 4)^2 \quad (5)$$

which reduces to a quadratic in x , viz.

$$x^2 - 10x + 16 = 0 \quad (6)$$

whose roots are 8 and 2. As both are positive log is defined for both. So it is tempting to think that $f'(x)$ has at least two roots because

of the first factor. But we recall here a word of caution given after the solution of Q.3 above regarding the conversion of one equation to another. In going from (3) to (5) we first applied exponentiation and then squaring. While exponentiation is a reversible operation, squaring is not so. (That is, $e^a = e^b$ implies $a = b$. But even though $a = b$ implies $a^2 = b^2$ the converse is not true.) So we must check whether the two solutions of (5) we got are also solutions of (4). For $x = 8$, both the sides of (4) are 8 each. But for $x = 2$, the L.H.S. is 6 while the R.H.S. is 2. So we have to discard $x = 2$. Hence $x = 8$ is the only solution to (4). (We need not check if it is also a solution of (3), because as already said the reduction from (3) to (4) is a reversible one.)

Let us now see if we get any other roots of $f'(x)$ from the last factor of the R.H.S. of (2). Cancelling $\sqrt{2}$ from the numerator and the denominator of the first term, the equation we get is

$$\frac{1}{2(4 + \sqrt{2x})\sqrt{x}} = \frac{1}{x} \quad (7)$$

or, taking reciprocals (which is a reversible operation, but which reverses inequalities),

$$2(4 + \sqrt{2x})\sqrt{x} = x \quad (8)$$

On expansion the L.H.S. equals

$$8\sqrt{x} + 2\sqrt{2x} \quad (9)$$

which is clearly bigger than x for all $x > 0$. So, we conclude that the last factor of the R.H.S. of (2) has no roots. In fact, it is always negative because in (8) the L.H.S. is bigger than the R.H.S. and so its reciprocal is smaller than that of the R.H.S. (An easier way to see that this factor is negative is to write $\frac{4 + \sqrt{2x}}{x}$ as $\frac{4}{x} + \frac{\sqrt{2}}{\sqrt{x}}$ and note that both the terms have negative derivatives for $x > 0$.)

Summing up, $x = 8$ is the only critical point of $f(x)$. We now check how the derivative $f'(x)$ changes its sign at 8. As already noted, the last factor of the R.H.S. of (2) is negative for all $x > 0$. To check the sign of the middle factor for $x < 8$ it suffices to give x any value between 0 and 8. (Normally, one takes points that are close to 8. But in the

present case we already know that $f'(x)$ has no other zeros and therefore it maintains its sign on the interval $(0, 8)$ by the Intermediate Value Property.) Taking $x = 2$, $4 + \sqrt{2x} = 6 > x$. So, $\log_e(4 + \sqrt{2x}) > \log_e x$ for $x = 2$ and hence for all $x \in (0, 8)$. (Here we are crucially using that the base of the logarithms, viz. e , is bigger than 1. As pointed out in the solution to Q.2, if the base is less than 1 then larger numbers have smaller logarithms.) To see what happens on the other side of 8, it again suffices to give x any value of our choice, We let $x = 32$. Then $4 + \sqrt{2x} = 12$ which is less than $x (=32)$. Thus the middle factor of the R.H.S. of (2) changes sign from $+$ to $-$ as x passes through 8. The first factor, viz. 2, is a positive constant. But the last factor is negative. As a result, the sign of $f'(x)$ changes from $-$ to $+$ as x passes through 8. So $f(x)$ decreases upto some point (specifically, upto, $x = 8$) and increases throughout after that. ■

There is also a simpler way of arriving at the answer. The given function $f(x)$ is $(u(x))^2$ where

$$u(x) = \log_e \left(\frac{4 + \sqrt{2x}}{x} \right) = \log_e(4 + \sqrt{2x}) - \log_e x \quad (10)$$

So we can study the behaviour of $f(x)$ by studying $u(x)$. We already saw that $u(x)$ vanishes only at $x = 8$. For $x < 8$, $4 + \sqrt{2x} > x$ and so $u(x) > 0$. But for $x > 8$, $4 + \sqrt{2x} < x$ and so $u(x)$ is negative. Further,

$$u'(x) = \frac{\frac{\sqrt{2}}{2}}{2(4 + \sqrt{2x})\sqrt{2x}} - \frac{1}{x} \quad (11)$$

which, as we already saw, is negative for all $x > 0$. So $u(x)$ is a strictly decreasing function on $(0, \infty)$. On $(0, 8)$, $u(x)$ is positive and decreasing. Hence so is its square. But on $(8, \infty)$, even though $u(x)$ is decreasing, it is negative and therefore its square is increasing!. So the correct option is (A). ■

The essential idea in the problem, viz. to determine the increasing/decreasing behaviour of a (differentiable) function by looking at the sign of its derivative, is very common. But the numerical data has been carefully designed to test if the candidate is careful in converting one equation to another, something which was not tested in Q.3.

Q.24 What is the number of ordered triplets (a, b, c) , where a, b, c are positive integers (not necessarily distinct), such that $abc = 1000$?

- (A) 64 (B) 100 (C) 200 (D) 560

Answer and Comments: (B). In essence, the question asks in how many ways 1000 can be factored as a product of three factors. Since $1000 = (10)^3 = 5^3 2^3$ each factor will have to be of the form $5^r 2^s$ for some integers, r, s with $0 \leq r \leq 3$ and $0 \leq s \leq 3$. So, if we let

$$a = 5^{r_1} 2^{s_1}, b = 5^{r_2} 2^{s_2} \text{ and } c = 5^{r_3} 2^{s_3} \quad (1)$$

then we have two equations

$$r_1 + r_2 + r_3 = 3 \quad (2)$$

$$\text{and } s_1 + s_2 + s_3 = 3 \quad (3)$$

We first count the number of triplets (r_1, r_2, r_3) of non-negative integers which add up to 3. In other words we find the number of elements in the set R defined by

$$R = \{(r_1, r_2, r_3) : 0 \leq r_i \leq 3 \text{ for } i = 1, 2, 3 \text{ and } r_1 + r_2 + r_3 = 3\} \quad (4)$$

Note that r_3 is determined by r_1 and r_2 . The possibilities for (r_1, r_2) are $(0, 0), (1, 0), (0, 1), (1, 1), (2, 0), (0, 2), (3, 0), (2, 1), (1, 2)$ and $(0, 3)$. Thus there are 10 such triplets. (Note that $10 = \binom{5}{2}$). This is not a coincidence. The problem is equivalent to partitioning a row of three dots into three parts by drawing two vertical walls to separate the parts. Thus the arrangements

$$\vdots \bullet \vdots \bullet \bullet \quad \text{and} \quad \vdots \bullet \bullet \bullet \vdots$$

represent the triplets $(0, 1, 2)$ and $(0, 3, 0)$ respectively. Three identical dots and two identical walls can be arranged in $\binom{5}{2}$ ways. More generally, a row of n dots can be partitioned into k parts in $\binom{n+k-1}{n}$ ways.)

Similarly, if we let

$$S = \{(s_1, s_2, s_3) : 0 \leq s_i \leq 3 \text{ for } i = 1, 2, 3 \text{ and } s_1 + s_2 + s_3 = 3\} \quad (5)$$

then S also has 10 triplets in it.

Now every factorisation abc of 1000 is of the form $(5^{r_1}2^{s_1})(5^{r_2}2^{s_2})(5^{s_3}2^{r_3})$ with $(r_1, r_2, r_3) \in R$ and $(s_1, s_2, s_3) \in S$. As each triplet can be chosen in 10 ways independently of the other, the number of possible factorisations is $10 \times 10 = 100$. ■

The number theory part of the problem is elementary. Thereafter it reduces to a combinatorial problem. As the exponents of 5 and 2 in the prime power factorisation of 1000 are the same (viz. 3 each), there is some duplication of work in calculating $|R|$ and $|S|$. If instead of 1000, the number given was 2000, the counts would have been different. But the method would be the same. The paper-setters probably realised that nothing new is involved and so to reduce the computations gave 1000 instead of 2000. Like many other problems, this one is a disguised combinatorial problem of counting the number of ways you can put n balls into k distinct boxes. Q.10 about chocolate distribution to students was also a problem of this type, but with the additional restrictions that no box be empty and further that exactly two boxes are to contain at least two balls. Having already asked that problem, there is not much point in asking another, superficially garbed as a number theory problem. Perhaps the limited syllabus of ISI entrance (in comparison to the JEE syllabus) forces such duplications.

Q.25 Let $f : (0, \infty) \rightarrow (0, \infty)$ be a function differentiable at 3, and satisfying $f(3) = 3f'(3) > 0$. Then the limit

$$\lim_{x \rightarrow \infty} \left(\frac{f\left(3 + \frac{3}{x}\right)}{f(3)} \right)^x$$

- | | |
|-----------------------------------|--------------------------------|
| (A) exists and is equal to 3 | (B) exists and is equal to e |
| (C) exists and is equal to $f(3)$ | (D) need not always exist. |

Answer and Comments: (B). The standard procedure to handle limits of exponential functions where both the base and the exponents are variable, is by taking logarithms. Apart from doing away with a monstrous notation, it also helps to evaluate the limit, because of the continuity of the logarithm function, or rather, that of its inverse, viz, the exponential function.

So, let $g(x)$ denote the given function. Then

$$\ln g(x) = x \ln \left[\frac{f\left(3 + \frac{3}{x}\right)}{f(3)} \right] \quad (1)$$

(Here $\ln$ stands for $\log_e$, i.e. the logarithm with base e . This notation is now getting increasingly common.) We rewrite this using the laws of logarithms as

$$\ln g(x) = x \left[\ln f\left(3 + \frac{3}{x}\right) - \ln f(3) \right] \quad (2)$$

(Note that there is no difficulty in taking logarithms, because $f(3)$ is given to be positive and continuous (in fact, differentiable) at 3 and hence will also be positive in a neighbourhood of 3 which will include $3 + \frac{3}{x}$ if $x > 0$ is sufficiently large.)

As $x \rightarrow \infty$, the first factor x in the R.H.S. of (2) tends to ∞ . But the second factor tends to 0 because $3 + \frac{1}{x}$ tends to 3 and f is continuous at 3. So this is a limit of an indeterminate form of the $\infty \times 0$ type. The l'Hôpital's rule is a standard tool to handle such limits. But it is more easily applicable to limits of the indeterminate forms of the $\frac{0}{0}$ type. This can be arranged by introducing a new variable $y = \frac{1}{x}$. Then $y \rightarrow 0^+$ as $x \rightarrow \infty$. So,

$$\lim_{x \rightarrow \infty} \ln g(x) = \lim_{y \rightarrow 0^+} \frac{\ln f(3 + 3y) - \ln f(3)}{y} \quad (3)$$

It is now tempting to apply l'Hôpital's rule as f is given to be differentiable at 3. But we cannot do so because l'Hôpital's rule requires differentiability of $\ln f(3 + 3y)$ not so much at 0 but in a (deleted) neighbourhood of 0. Equivalently we need the differentiability of f not

so much at 3 but in a neighbourhood of 3. And that is not given or implied by the data. But we can apply what is given, viz. that f and hence $\ln f$ is differentiable at 3. Call $3y$ as z . Then as $y \rightarrow 0^+$, so does z . So, continuing the reduction above,

$$\begin{aligned}\lim_{x \rightarrow \infty} \ln g(x) &= \lim_{z \rightarrow 0^+} \frac{3(\ln f(3+z) - \ln f(3))}{z} \\ &= 3 \frac{f'(3)}{f(3)}\end{aligned}\tag{4}$$

from the chain rule and the very definition of derivative. The data gives this ratio as 1. So $\ln g(x) \rightarrow 1$ as $x \rightarrow \infty$ and therefore the original limit $\lim_{x \rightarrow \infty} g(x)$ equals e^1 , i.e. e . ■

The paper setters have been careful to give $f(3) > 0$ to ensure that $\ln f(3)$ is defined. Even if $f(3)$ were negative, by continuity of f , $f(3+\frac{1}{x})$ would be negative for sufficiently large x . So the ratio $\frac{f(3+\frac{1}{x})}{f(3)}$ is positive and its log makes sense. But it cannot be expressed in terms of the logs of the numerator and the denominator without resorting to complex numbers. The way out when $f(3) < 0$ is to replace f by $-f$. Unfortunately, such finesse is likely to be lost on most candidates. In fact, many of them might not even notice the inapplicability of the l'Hôpital's rule. They will simply differentiate the numerator of the R.H.S. of (3) to equate the limit with $\lim_{y \rightarrow 0^+} \frac{3f'(3+3y)}{f(3+3y)}$ and conclude

that this limit is $\frac{3f'(3)}{f(3)}$ which equals 1 from the data. But this is the fate of so many MCQ's. You can get the right answer by questionable means. Only a conventional examination can distinguish the scrupulous from the sloppy. This distinction may not be so vital for JEE, which is, after all, an entrance test for an engineering program. But the ISI tests are meant for a degree program in Mathematics or Statistics. So such a question will pass the muster for JEE but not the present test.

Q.26 Let z be a non-zero complex number such that $\left|z - \frac{1}{z}\right| = 2$. What is the maximum value of $|z|$?

- (A) 1 (B) $\sqrt{2}$ (C) 2 (D) $1 + \sqrt{2}$

Answer and Comments: (D). Since the question refers to $\frac{1}{z}$ we have to assume $z \neq 0$. So we can take z in the polar form. That is,

$$z = r(\cos \theta + i \sin \theta) = re^{i\theta} \quad (1)$$

where $|z| = r$. Then

$$\frac{1}{z} = \frac{1}{r}e^{-i\theta} = \frac{1}{r}(\cos \theta - i \sin \theta) \quad (2)$$

So, by a direct calculation

$$\begin{aligned} \left| z - \frac{1}{z} \right|^2 &= (z - \frac{1}{z})(\bar{z} - \frac{1}{\bar{z}}) \\ &= |z|^2 + \frac{1}{|z|^2} - \frac{z}{\bar{z}} - \frac{\bar{z}}{z} \\ &= r^2 + \frac{1}{r^2} - e^{2i\theta} - e^{-2i\theta} \\ &= r^2 + \frac{1}{r^2} - 2 \cos 2\theta \end{aligned} \quad (3)$$

We are given that $\left| z - \frac{1}{z} \right| = 2$. So we get an equation

$$r^2 + \frac{1}{r^2} - 2 \cos 2\theta - 4 = 0 \quad (4)$$

Rewriting this as a quadratic in r^2 , we get

$$r^4 - (4 + 2 \cos 2\theta)r^2 + 1 = 0 \quad (5)$$

This gives

$$r^2 = 2 + \cos 2\theta \pm \sqrt{(2 + \cos 2\theta)^2 - 1} \quad (6)$$

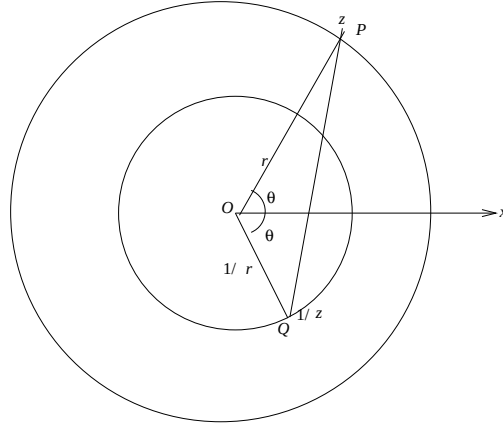
As we are interested in maximising r (and hence r^2) we discard the negative square root and get

$$r^2 = 2 + \cos 2\theta + \sqrt{(2 + \cos 2\theta)^2 - 1} \quad (7)$$

which will be maximum when $\cos 2\theta = 1$. The maximum value of r^2 is $3 + \sqrt{8} = 3 + 2\sqrt{2}$. We recognise this as $(1 + \sqrt{2})^2$. So the maximum value of r , i.e. of $|z|$ is $1 + \sqrt{2}$. ■

Although not asked in the problem, we can also locate the points where the maximum occurs. They are the ones where $\cos 2\theta = 1$, i.e. $\theta = 0$ or $\theta = \pi$. The first value gives $z = 1 + \sqrt{2}$ and the second, $z = -1 - \sqrt{2}$. Note that both lie on the real axis.

Actually, (3) could also have been derived geometrically as we now show.



Let the points P and Q in the Argand diagram represent the complex numbers z and $\frac{1}{z}$ respectively. (In the diagram above, it is assumed that $|z| > 1$. That is why P lies on the outer and Q on the inner circle of radii r and $\frac{1}{r}$ respectively. If $|z| < 1$, it will be the other way round.) Note that both OP and OQ make angle θ with the x -axis but lie on opposite sides of it.

The lengths of the segments OP, OQ and PQ are, $r(=|z|)$, $\frac{1}{r}$ and $\left|z - \frac{1}{z}\right|$ respectively. Also $\angle POQ = 2\theta$. So applying the cosine rule to the triangle OPQ we get

$$\left|z - \frac{1}{z}\right|^2 = r^2 + \frac{1}{r^2} - 2r \times \frac{1}{r} \cos 2\theta = r^2 + \frac{1}{r^2} - 2 \cos 2\theta \quad (8)$$

which is exactly (3).

We can also get the answer by the triangle inequality which gives $|z| \leq \left|z - \frac{1}{z}\right| + \frac{1}{|z|} = 2 + \frac{1}{|z|}$. This reduces to $(|z| - 1)^2 \leq 2$. So $|z|$ lies between $1 - \sqrt{2}$ and $1 + \sqrt{2}$.

Interestingly, there is also a solution using a less known form of the triangle inequality. Normally, it is expressed as

$$|z_1 + z_2| \leq |z_1| + |z_2| \quad (9)$$

and the geometric interpretation is that the sum of the two sides of any triangle exceeds the third, with equality only for a degenerate triangle. Another, but less well known version of this result is that the difference of any two sides is less than the third side. So for complex numbers the triangle inequality in this form would read

$$||z_1| - |z_2|| \leq |z_1 - z_2| \quad (10)$$

To prove this, assume, without loss of generality that $|z_1| \geq |z_2|$. Then the L.H.S. is $|z_1| - |z_2|$ and the inequality follows by applying the usual form of the triangle inequality to the sum $(z_1 - z_2) + z_2$.

We apply (10) taking z_1 and $z_2 = \frac{1}{z}$. Then the given condition means

$$||z| - \frac{1}{|z|}| \leq 2 \quad (11)$$

Squaring and calling $|z|$ as r , we get $(r^2 - 1)^2 \leq 4r^2$ which can be written as

$$r^4 - 6r^2 - 1 \leq 0 \quad (12)$$

Treating the L.H.S. as a quadratic in r^2 , we see that r^2 must lie in between the two roots of the quadratic, which are $3 \pm 2\sqrt{2}$. As $r^2 > 0$, the maximum value it can have is $3 + 2\sqrt{2}$. Recognising this as $(1 + \sqrt{2})^2$, we get $r \leq 1 + \sqrt{2}$.

It would be difficult to attempt the problem by taking z in the usual cartesian form as $x + iy$ where x, y are real. Keeping in mind that $\frac{1}{z} = \frac{\bar{z}}{|z|^2}$, the condition $\left|z - \frac{1}{z}\right| = 2$ would then translate into

$$\left|x + iy - \frac{x - iy}{x^2 + y^2}\right|^2 = 4 \quad (13)$$

which will take a very complicated form when expressed in terms of x and y . So even through the function to be maximised here, viz.

$x^2 + y^2$, is relatively easy, the constraint on x and y imposed by (8) is prohibitively complicated even for those who are familiar with what is called **constrained optimisation** i.e. maximising or minimising a function of two variables, subject to a constraint. (In Q.6 of Paper 2 below the function $f(x, y) = x^2 + y^2$ is maximised on the ellipse $\{(x, y) : x^2 - 2xy + y^2 = 1\}$, by parametrising the ellipse in terms of a certain parameter θ . But such a parametrisation is not easy for the curve represented by (8).)

The moral of the story is that just because one is familiar with some method should not lead one to think that it is the golden method which will be applicable everywhere. It is equally important to realise (and realise *fast*) where it will *not* be applicable. In the present problem, the polar representation of a complex number worked.

Q.27 The minimum value of

$$|\sin x + \cos x + \tan x + \operatorname{cosec} x + \sec x + \tan x| \text{ is}$$

$$(A) 0 \quad (B) 2\sqrt{2} - 1 \quad (C) 2\sqrt{2} + 1 \quad (D) 6$$

Answer and Comments: (B). Define

$$f(x) = \sin x + \cos x + \tan x + \cot x + \sec x + \operatorname{cosec} x \quad (1)$$

Our task is to minimise $|f(x)|$. (Note that we have changed the order of the terms slightly, because we are more used to study $\tan$ and $\cot$ together as their properties are similar and they are intimately related by $\cot x = \tan(\frac{\pi}{2} - x)$. Also we have many simple identities such as

$$\begin{aligned} \tan x + \cot x &= \frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} \\ &= \frac{1}{\sin x \cos x} \end{aligned} \quad (1)$$

$$= \frac{2}{\sin 2x} \quad (2)$$

and a charming wealth of many more results. The same holds for the pair $\sec x$ and $\operatorname{cosec} x$. It is not clear why the paper setters have listed

the terms in an ‘unnatural’ order, not that it makes any difference mathematically.)

Anyway coming to the problem, the function $f(x)$ is not defined when x is a multiple of $\frac{\pi}{2}$. But as x tends to any such point, $f(x) \rightarrow \pm\infty$, and so $|f(x)| \rightarrow \infty$. So, such points do not matter when the task is to minimise, rather than maximise $|f(x)|$. We exclude them from all our discussion. Also, because of the periodicity of all the six terms, it suffices to take only the values of x in $(0, 2\pi)$. We shall frequently have to divide this domain into four parts that lie in the four quadrants, (i.e. the intervals, $(0, \pi/2)$, $(\pi/2, \pi)$, $(\pi, 3\pi/2)$ and $(3\pi/2, 2\pi)$.)

Let us first dismiss option (A). This is not as simple as it appears. It is equivalent to showing that $f(x)$ has no zeros in $(0, 2\pi)$. This is obvious when x is in the first quadrant where all the six trigonometric functions are positive. But in each of the remaining three quadrants, only two of them are positive and the remaining four negative. So to prove that f has no zeros in any of them takes some work. We first simplify the expression for $f(x)$ so that it involves only the two basic functions $\sin x$ and $\cos x$ with which we are most familiar.

$$\begin{aligned} f(x) &= \sin x + \cos x + \frac{1 + \sin x}{\cos x} + \frac{1 + \cos x}{\sin x} \\ &= \frac{(\sin x + \cos x) \sin x \cos x + \sin x + \cos x + 1}{\sin x \cos x} \\ &= \frac{(\sin x + \cos x)(\sin x \cos x + 1) + 1}{\sin x \cos x} \end{aligned} \tag{3}$$

$$= \frac{\sqrt{2} \sin(x + \frac{\pi}{4})(1 + \frac{1}{2} \sin 2x) + 1}{\sin x \cos x} \tag{4}$$

Hence the equation $f(x) = 0$ reduces to

$$\sqrt{2} \sin(x + \frac{\pi}{4})(1 + \frac{1}{2} \sin 2x) = -1 \tag{5}$$

(Note, however, that the reduction is not reversible. There are solutions of (5) which are not solutions of $f(x) = 0$. For example, $x = \pi/2$ and $x = \pi$ are solutions of (5). But $f(x)$ is not defined at these values of x and so they are not solutions of $f(x) = 0$.)

We now analyse (5). The second factor of the L.H.S. is always non-negative. So the first factor has to be negative. That puts $x + \frac{\pi}{4}$ in the

interval $(\pi, 2\pi)$ and thus gives

$$\frac{3\pi}{4} < x < \frac{7\pi}{4} \quad (6)$$

This interval cuts across three quadrants, the part $(3\pi/4, \pi)$ in the second quadrant, the part $(\pi, 3\pi/2)$ in the third quadrant and the part $(3\pi/2, 7\pi/4)$ in the fourth quadrant. We prove the impossibility of (5) in each of them separately. The reasoning is essentially the same in all three. We do the case of the interval $(\pi, 3\pi/2)$ and leave the remaining as exercises (a convenient escape for authors!).

So suppose that $x \in (\pi, 3\pi/2)$. Then x is in the third quadrant and so $\sin x, \cos x$ are both negative. Put $y = \pi - x$. Then y will be in the first quadrant $(0, \pi/2)$. Also

$$\begin{aligned} \sin(x + \frac{\pi}{4}) &= \sin(y + \pi + \frac{\pi}{4}) = -\sin(y + \frac{\pi}{4}) \\ \text{and } 1 + \sin 2x &= 1 + \sin(2y + 2\pi) = \sin 2y \end{aligned} \quad (7)$$

As a result, (4) is transformed to

$$\sqrt{2} \sin(y + \frac{\pi}{4})(1 + \frac{1}{2} \sin 2y) = 1 \quad (8)$$

We want to show that this equation has no solution for $y \in (0, \pi/2)$. Call the L.H.S. as $h(y)$. Then $h(y)$ is symmetric about $\frac{\pi}{4}$, i.e. $h(\frac{\pi}{2} - y) = h(y)$ for all $y \in (0, \pi/2)$. (This becomes more evident if we rewrite $h(y)$ in terms of $\sin y$ and $\cos y$ and use $\sin(\frac{\pi}{2} - y) = \cos y$ and $\cos(\frac{\pi}{2} - y) = \sin y$.) So it suffices to show that $h(y) = 1$ has no solution on $(0, \pi/4]$. Of course, $y = 0$ is a solution but we are discarding it. Now as y increases from 0 to $\pi/4$, both the factors $\sin(y + \frac{\pi}{4})$ and $(1 + \frac{1}{2} \sin 2y)$ are strictly increasing. So $h(y) > h(0) = 1$ for $y \in (0, \pi/4]$. Hence (1) cannot hold for $y \in (0, \pi/4]$.

So, at long last we have proved that $f(x)$ can never vanish and therefore the minimum of $|f(x)|$ is positive, ruling out the option (A). We now proceed to determine this minimum using calculus. Minimising $|f(x)|$ is equivalent to minimising $|f(x)|^2$ which is the same as $(f(x))^2$. So, we let

$$g(x) = (f(x))^2 = (\sin x + \cos x + \tan x + \cot x + \sec x + \operatorname{cosec} x)^2 \quad (9)$$

(The reason we choose $g(x)$ over $|f(x)|$ is the non-differentiability of the absolute value function.) By a direct calculation,

$$g'(x) = 2f(x)f'(x) \quad (10)$$

We just showed that the first factor never vanishes. So, we look for the zeros of $f'(x)$ to locate the possible points of minimum of $g(x)$. (As already noted, $g(x)$ has no maximum.)

We calculate $f'(x)$ and simplify it to express it solely in terms of $\sin x$ and $\cos x$.

$$\begin{aligned} f'(x) &= \cos x - \sin x + \sec^2 x - \cot^2 x + \sec x \tan x - \operatorname{cosec} x \cot x \\ &= \cos x - \sin x + \frac{1}{\cos^2 x} - \frac{1}{\sin^2 x} + \frac{\sin x}{\cos^2 x} - \frac{\cos x}{\sin^2 x} \\ &= \cos x - \sin x + \frac{\sin^2 x - \cos^2 x}{\sin^2 x \cos^2 x} + \frac{\sin^3 x - \cos^3 x}{\sin^2 x \cos^2 x} \end{aligned} \quad (11)$$

So, the equation $f'(x) = 0$ reduces to

$$(\sin^2 x - \cos^2 x) + (\sin^3 x - \cos^3 x) = (\sin x - \cos x) \sin^2 x \cos^2 x \quad (12)$$

Both the sides have $\sin x - \cos x$ as a factor. So one possibility is

$$\sin x = \cos x \quad (13)$$

which gives $\pi/4$ and $3\pi/4$ as critical points. The other possibility is

$$(\sin x + \cos x) + (1 + \sin x \cos x) = \sin^2 x \cos^2 x \quad (14)$$

Factorising both the sides,

$$(1 + \sin x)(1 + \cos x) = (1 + \cos x)(1 - \cos x)(1 + \sin x)(1 - \sin x) \quad (15)$$

The factors $1 + \sin x$ and $1 + \cos x$ are both non-zero as we are not allowing x to take multiples of $\pi/2$ as values. So (15) simplifies to

$$\sin x + \cos x = \sin x \cos x \quad (16)$$

Squaring both the sides

$$1 + \sin 2x = \frac{1}{4} \sin^2 2x \quad (17)$$

which is a quadratic in $\sin x$, viz.

$$\sin^2 2x - 4 \sin 2x - 4 = 0 \quad (18)$$

whose roots are $2 \pm 2\sqrt{2}$. Since $\sin 2x$ cannot exceed 1, we choose the negative square root and get

$$\sin 2x = 2 - 2\sqrt{2} \quad (19)$$

There are two values of x in $(0, 2\pi)$ which satisfy this equation. We can identify them explicitly (with a calculator, their degree measures are approximately 118° and 332° , symmetrically located about the line $y = x$ as they ought to). But that is hardly necessary because the value of f at both of them is the same and can be found as follows. From (19),

$$\sin x \cos x = 1 - \sqrt{2} \quad (20)$$

Combining this with (16), we have

$$\sin x + \cos x = 1 - \sqrt{2} \quad (21)$$

$f(x)$ can be evaluated from these two equations.

$$\begin{aligned} |f(x)| &= |(2 - \sqrt{2}) + \frac{1}{1 - \sqrt{2}}| \\ &= |(2 - \sqrt{2}) - (1 + \sqrt{2})| \\ &= |1 - 2\sqrt{2}| \\ &= 2\sqrt{2} - 1 \end{aligned} \quad (22)$$

since $1 < 2\sqrt{2}$. So, we now have four critical points of $f(x)$, $\frac{\pi}{4}$ and $\frac{3\pi}{4}$ obtained from (13) and the two unidentified solutions of (19) at each of which $|f(x)| = 2\sqrt{2} - 1$. We must compare this value with $|f(\pi/4)|$ and $|f(3\pi/4)|$ to decide which is the smallest.

$$|f(\frac{\pi}{4})| = \sqrt{2} + 2 + 2\sqrt{2} = 2 + 3\sqrt{2} \quad (23)$$

while

$$|f(\frac{3\pi}{4})| = |-\sqrt{2} + 2 - 2\sqrt{2}| = 3\sqrt{2} - 2 \quad (24)$$

Among (22), (23) and (24), clearly (22) is the winner since $\sqrt{2} > 1$. ■

The symmetry of $f(x)$ about the point $\frac{\pi}{4}$ (i.e. the fact that $f(x) = f(\frac{\pi}{2} - x)$) can be used advantageously to simplify the calculations with a suitable substitution. Put $u = \sin x + \cos x = \sqrt{2} \cos(\frac{\pi}{4} - x)$. Keeping in mind that $2 \sin x \cos x = (\sin x + \cos x)^2 - 1 = u^2 - 1$, the R.H.S. of (3) gives a new function $h(u)$ of u defined by

$$h(u) = \frac{u(u^2 + 1) + 2}{u^2 - 1} \quad (25)$$

Note that $u \neq -1$ as otherwise x takes a forbidden value. So, (25) simplifies as

$$h(u) = \frac{u^3 + 1 + u + 1}{u^2 - 1} = \frac{u^2 - u + 2}{u - 1} = u + \frac{2}{u - 1} \quad (26)$$

It is much easier to minimise $|h(u)|$ as u varies over $[-\sqrt{2}, \sqrt{2}]$.

A very interesting problem because the function to be minimised is unusually intimidating, being a sum of no less than six functions. Because of the A.M.-G.M. inequality, the minimum of a sum of a number and its reciprocal is 2. Applying this to the three pairs of reciprocals in the six trigonometric functions, it is tempting to think that the minimum is 6. But this turns out not to be the case. $\pi/4$ and $3\pi/4$ are also tempting options for the occurrence of the minimum. But the correct critical point is far from obvious and the search for it involves considerable trigonometric algebraic manipulations. Perhaps, as in Q.5, the paper setters intended to reward those who can take a hint and think of the substitution $u = \sin x + \cos x$. (The problem seems to be picked up from the 2003 Putnam competition. There a candidate had 30 minutes to solve it. Here he has only 4. This is insulting to a good problem.)

Q.28 For any function $f : X \longrightarrow Y$ and any subset A of Y , define

$$f^{-1}(A) = \{x \in X : f(x) \in A\}.$$

Let A^c denote the complement of A in Y . For subsets A_1, A_2 of Y , consider the following statements:

$$(i) \quad f^{-1}(A_1^c \cap A_2^c) = (f^{-1}(A_1))^c \cup (f^{-1}(A_2))^c$$

(ii) If $f^{-1}(A_1) = f^{-1}(A_2)$ then $A_1 = A_2$.

Then,

- (A) both (i) and (ii) are always true
- (B) (i) is always true, but (ii) may not always be true
- (C) (ii) is always true, but (i) may not always be true
- (D) neither (i) nor (ii) is always true.

Answer and Comments: (D). For a subset A of Y , the set A^c is defined in the question as the complement of A in Y , also often denoted by $Y - A$. But statement (i) in the question makes a reference to the sets $(f^{-1}(A_1))^c$ and $(f^{-1}(A_2))^c$. The definition given is inapplicable here because $f^{-1}(A_1)$ and $f^{-1}(A_2)$ are subsets of X rather than of Y . But, by analogy, we take them to mean the complements of $f^{-1}(A_1)$ and $f^{-1}(A_2)$ in X . That is $X - f^{-1}(A_1)$ and $X - f^{-1}(A_2)$ respectively.

With this understanding it is very easy to disprove (ii). As a simple counterexample, let X be any non-empty set and $Y = \{1, 2, 3\}$ (or any other set with three or more elements). Define $f : X \rightarrow Y$ by $f(x) = 1$ for all $x \in X$. In other words, f is the constant function 1. Take $A_1 = \{2\}$ and $A_2 = \{3\}$. Then $A_1 \neq A_2$. But $f^{-1}(A_1)$ and $f^{-1}(A_2)$ are both empty sets and hence equal.

To decide if Statement (i) is true we need the following three simple properties of the inverse f^{-1} operator. (We call it ‘operator’ and not ‘function’ because f^{-1} acts on subsets of Y and not on the elements of Y as the inverse function, if it exists, does. The inverse function exists only when f is a bijection, i.e. both one-to-one and onto. But that is not given to be the case. For the operator f^{-1} there is no restriction on f .) These properties are immediate from the very definition. Verbally, they assert that the inverse operator preserves the union, the intersection and the complements of subsets of Y . That is, for every two subsets A_1, A_2 of Y , we have

$$f^{-1}(A_1 \cup A_2) = f^{-1}(A_1) \cup f^{-1}(A_2) \quad (1)$$

$$f^{-1}(A_1 \cap A_2) = f^{-1}(A_1) \cap f^{-1}(A_2) \quad (2)$$

$$\text{and } f^{-1}(A_1^c) = (f^{-1}(A_1))^c \quad (3)$$

The proofs are straightforward, well-known and hence omitted.

Now, by the second and the third properties,

$$f^{-1}(A_1^c \cap A_2^c) = f^{-1}(A_1^c) \cap f^{-1}(A_2^c) = (f^{-1}(A_1))^c \cap (f^{-1}(A_2))^c \quad (4)$$

There is no reason why the R.H.S. should equal $(f^{-1}(A_1))^c \cup (f^{-1}(A_2))^c$ as claimed by (i). In general the intersection of two subsets is much smaller than their union. In fact it could even be empty even though neither set is empty. Again, for an easy counterexample, let X be any non-empty set. Take $Y = X$ and f to be the identity function. Then $f^{-1}(A) = A$ for all $A \subset Y$ and so (i) will fail when $A_1 = \emptyset$ and $A_2 = Y$. So, both (i) and (ii) are false in general. ■

Apparently, (i) is designed to fool the candidates who vaguely remember what are called **DeMorgan's laws** about complements. Verbally, they assert that the complement of the union is the intersection of the complements and the complement of an intersection is the union of the complements. In symbols, for every two subsets B_1, B_2 of a set X ,

$$(B_1 \cup B_2)^c = B_1^c \cap B_2^c \quad (5)$$

$$\text{and } (B_1 \cap B_2)^c = B_1^c \cup B_2^c \quad (6)$$

Because of the second law, we have

$$A_1^c \cap A_2^c = (A_1 \cup A_2)^c \quad (7)$$

and hence by (1) to (3),

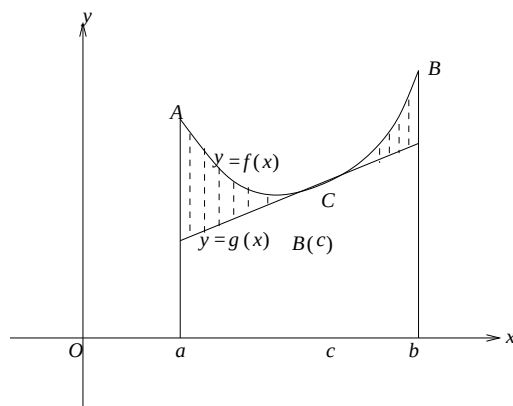
$$\begin{aligned} f^{-1}(A_1^c \cap A_2^c) &= f^{-1}((A_1 \cup A_2)^c) \\ &= (f^{-1}(A_1 \cup A_2))^c \\ &= (f^{-1}(A_1) \cup f^{-1}(A_2))^c \end{aligned} \quad (8)$$

If we apply DeMorgan's laws to the R.H.S., we would get $(f^{-1}(A_1))^c \cap (f^{-1}(A_2))^c$ which we obtained earlier without DeMorgan's laws. So DeMorgan's laws are not needed in this problem. They are surely useful elsewhere. But vaguely remembered laws are a liability rather than an asset. And, in any case, memory, even when correct, is no substitute for logical reasoning. All it does is to save some time.

Q.29 Let f be a function such that $f''(x)$ exists, and $f''(x) > 0$ for all $x \in [a, b]$. For any point $c \in [a, b]$, let $A(c)$ denote the area of the region bounded by $y = f(x)$, the tangent to the graph of f at $x = c$ and the lines $x = a$ and $x = b$. Then

- (A) $A(c)$ attains its minimum at $c = \frac{1}{2}(a + b)$ for any such f
- (B) $A(c)$ attains its maximum at $c = \frac{1}{2}(a + b)$ for any such f
- (C) $A(c)$ attains its minimum at both $c = a$ and $c = b$ for any such f
- (D) the points c where $A(c)$ attains its minimum depend on f .

Answer and Comments: (A). The data implies that the function $f(x)$ is concave upwards on $[a, b]$. A well known characterisation of this (in fact, often taken as the very definition) is that for any two points $A = (a, f(a))$ and $B = (b, f(b))$ on the graph $y = f(x)$, the chord joining A and B lies entirely above the graph. Comparatively, a less known consequence of concavity upwards is that at any point $C = (c, f(c))$ on the graph, the tangent lies below the graph. (A proof is given after the solution.) So the data can be pictured as in the diagram below.



The problem asks for the maximisation/minimisation of the shaded area $A(c)$ as c varies from a to b . It is defined as

$$A(c) = \int_a^b (f(x) - g(x)) dx \quad (1)$$

where $y = g(x)$ is the equation of the tangent to the graph at the point $C = (c, f(c))$. Clearly the function $g(x)$ is defined by

$$g(x) = f(c) + f'(c)(x - a) \quad (2)$$

Putting this into (1)

$$A(c) = \int_a^b f(x) - f(c) - f'(c)(x - c) \, dx \quad (3)$$

Clearly,

$$A(c) = I - B(c) \quad (4)$$

where we let

$$I = \int_a^b f(x) \, dx \quad (5)$$

$$\text{and } B(c) = \int_a^b f(c) + f'(c)(x - c) \, dx \quad (6)$$

Since I is a constant independent of c , maximisation/ minimisation of $A(c)$ is equivalent to minimisation/maximisation of $B(c)$. Geometrically, $B(c)$ is the area of the trapezium with parallel sides along the lines $x = a$ and $x = b$. The perpendicular distance between these two lines is $b - a$. So,

$$\begin{aligned} B(c) &= \frac{b-a}{2}(g(a) + g(b)) \\ &= \frac{b-a}{2}[f(c) + f'(c)(a - c) + f(c) + f'(c)(b - c)] \\ &= \frac{b-a}{2}[2f(c) + f'(c)(a + b - 2c)] \end{aligned} \quad (7)$$

For fixed a, b and $f(x)$, $B(c)$ is a function of c . Differentiating w.r.t. c ,

$$\begin{aligned} B'(c) &= \frac{b-a}{2}[2f'(c) + f''(c)(a + b - 2c) - 2f'(c)] \\ &= \frac{b-a}{2}f''(c)(a + b - 2c) \end{aligned} \quad (9)$$

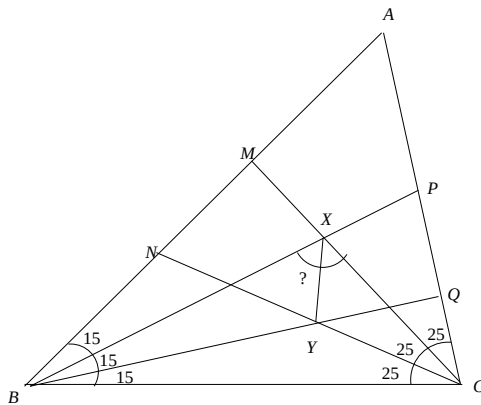
As $f''(c) > 0$, $B'(c) = 0$ implies that $c = \frac{1}{2}(a + b)$. Moreover, for $c_1 < \frac{1}{2}(a + b)$, $B'(c_1) = \frac{b-a}{2}f''(c_1)(a + b - 2c_1) > 0$ because f'' is given

to be positive everywhere. Similarly, for $c_2 > \frac{1}{2}(a+b)$, $B'(c_2) < 0$. So $B(c)$ changes its behaviour from increasing to decreasing at $\frac{1}{2}(a+b)$. Therefore, $B(c)$ attains its maximum at $c = \frac{1}{2}(a+b)$. As already noted, this shows that $A(c)$ attains its maximum at $c = \frac{1}{2}(a+b)$. ■

An interesting problem whose solution depends on the fact that if $f''(x) > 0$ for all $x \in [a, b]$, then for any $c \in [a, b]$, the graph lies above the tangent at the point $C = (c, f(c))$, i.e. $f(x) \geq g(x)$ with equality holding only at c . This can be proved using Lagrange's Mean Value Theorem which, when applied to f' , implies that f' is strictly increasing. So for $x > c$, by LMVT again, there is some $c_1 \in (c, x)$ such that $f(x) - f(c) = (x - c)f'(c_1) > (x - c)f'(c)$. So, $f(x) > f(c) + f'(c)(x - c) = g(x)$ for $x > c$. If $x < c$, by LMVT, $f(x) - f(c) = f'(c_2)(x - c)$ for some $c_2 \in (x, c)$. Since $f'(c) > f'(c_2)$ and $x - c < 0$, we have $f'(c_2)(x - c) > f'(c)(x - c)$, whence $f(x) > g(x)$.

The question has little to do with integration. It would fit more appropriately in Paper 2 where a candidate has to give justifications.

- Q.30 In $\triangle ABC$, the lines BP, BQ trisect $\angle ABC$ and the lines CM, CN trisect $\angle ACB$. Let BP and CM intersect at X and BQ and CN intersect at Y . If $\angle ABC = 45^\circ$ and $\angle ACB = 75^\circ$, then $\angle BXY$ is



- (A) 45° (A) $47\frac{1}{2}^\circ$ (C) 50° (D) 55°

Answer and Comments: (C). The paper setters have been kind enough to provide a diagram which makes it easy to understand a

problem. This is a relatively recent practice at least in mathematics. Traditionally, understanding a geometry problem correctly from its verbal statement was considered as a part and parcel of the problem. But complaints were raised from candidates who had had their education in some medium other than English. Technical terms like ‘orthocentre’ are not used in day to day life and so even a student with an otherwise excellent command over English may get stuck right at the start if a geometry problem involves this term, unless he has studied geometry in English. It would be unfair indeed if a candidate loses not because of poor knowledge of geometry but simply because he studied it in a different language. So this practice was started. Drawing a diagram also sometimes clears a possible ambiguity in a verbal statement. For example, given merely that two circles touch each other, whether they do so externally or internally. (See Q.22 above for an example of a geometry problem where not only no diagram is given, but no notations are given either. That is the way it used to be.)

But in the present problem, the paper setters have been too kind! That is, they have given the full diagram when only a part of it is needed for the solution. In the diagram given, simply concentrate on the triangle XBC . Then the point Y is simply the point of intersection of the angle bisectors of the angles $\angle XBC$ and $\angle XCB$. Since the three angle bisectors of any triangle are concurrent, the line XY bisects the angle $\angle BXC$. It only remains to find the measure of this angle. This is easy because the other two angles of the triangle XBC are 30° and 50° . So $\angle BXC = 180 - 30 - 50 = 100$ degrees. Hence $\angle BXY$ which is half of it is 50° . ■

The problem is simple once you focus on the essential part. No ingenious constructions, no complicated derivations are needed. A very good problem for an MCQ.

The angle trisectors of a triangle rarely figure in geometry. But there is a famous theorem due to **Morley** about them. Suppose that the angle trisectors of the third angle A of the triangle ABC (not shown in the figure) meet BP and CM at Z and W respectively. Then the triangle YZW is always equilateral no matter what the original triangle is. The proof is far from trivial.

Problems asking for the numerical measure of some angle, given those

of some other angles can often be quite tricky. There are a few problems where a trigonometric solution was known but a pure geometry solution was found only later. And there are a few people who spend hours and hours to find such a solution, even though the result *per se* is not new. That is the love of pure geometry, and, more generally of mathematics.

ISI BStat-BMath-UGB-2014 Paper with Comments

- Q. 1** A class has 100 students. Let $a_i, 1 \leq i \leq 100$ denote the number of friends the i -th student has in the class. For each $0 \leq j \leq 99$, let c_j denote the number of students having *at least* j friends. Show that:

$$\sum_{i=1}^{100} a_i = \sum_{j=1}^{99} c_j.$$

Solution: We begin by giving a ‘friendly’ interpretation of the sum $\sum_{i=1}^{100} a_i$. (Although not explicitly stated, it is inherent in the statement of the problem that the students are numbered from 1 to 100, as otherwise the expression i -th student makes no sense. We also assume that friendship is an irreflexive and symmetric relation, that is, no student is a friend of himself and further if x is a friend of y then y is a friend of x . Such assumptions are to be understood from common sense. Technically, the statement of the question should specify them. But then the statement would be very long and clumsy and sometimes ridiculous. That is why, in combinatorial problems where guests are to be seated on chairs, nobody stipulates that no two guests can occupy the same chair at the same time even though this is perfectly common in real life, especially in crowded vehicles!)

Now, coming to the problem, label the students as $v_1, v_2, \dots, v_{100}$. (We could use any other symbol instead of v . The reason for this particular choice will be clearer later.) a_i is the number of friends v_i has. Each such friend, say v_j , will have v_i as his friend. So the friendship between v_i and v_j will be counted twice, once in a_i and once in a_j . So, if we let F be the set of all friendly pairs, then

$$\sum_{i=1}^{100} a_i = 2|F| \tag{1}$$

where $|F|$ denotes the number of friendly pairs. (It is a good idea to verify such statements in some special cases just to ensure that no obvious mistake is committed. For example, if there are no friendships in the class then $a_i = 0$ for all i whence the L.H.S. is 0. And so is the

R.H.S. since in this case F is the empty set. At the other extreme, if every two (distinct) students are friends of each other, then in all there are $\binom{100}{2}$ elements in F and so the R.H.S. is 100×99 , which is also the L.H.S. since $a_i = 99$ for every i . As a medium example, if the 100 students are paired off into 50 mutually disjoint friendly pairs, then $|F| = 50$ and $a_i = 1$ for every i . So both the sides of (1) equal 100. Verification of such special cases is, of course, for your own assurance. It is best done mentally. Writing it out elaborately will only waste your time. You are not, after all, teaching the examiner!)

We now show that the number of friendly pairs can also be counted in another way. Classify the students according to how friendly they are. A student can have anywhere between 0 to 99 friends. For $0 \leq j \leq 99$, let T_j be the set of those students who have exactly j friends and denote $|T_j|$ by b_j . Each student in T_j has j friendly pairs, i.e. elements of F . Moreover, for a fixed j , all these jb_j pairs are distinct. (If two distinct students x and y in T_j have a common friend z , the pair $\{x, z\}$ is different from the pair $\{y, z\}$.) So the students in T_j contribute jb_j to F . But once again, every pair is counted twice. So we have

$$\sum_{j=0}^{99} jb_j = 2|F| \quad (2)$$

(Once again, for our own consolation, we verify this when there are 50 mutually disjoint friendly pairs. In this case $b_1 = 100$ and $b_j = 0$ for all $j \neq 1$. So both the sides of (2) equal 100. Similarly, in the extreme case where every two students are friendly, the R.H.S. is 9900 and so is the L.H.S. because now $b_{99} = 100$ and $b_j = 0$ for all $j \neq 99$.)

Putting (1) and (2) together we get

$$\sum_{i=1}^{100} a_i = \sum_{j=0}^{99} jb_j \quad (3)$$

So, to complete the solution we need to show that

$$\sum_{j=0}^{99} jb_j = \sum_{j=1}^{99} c_j \quad (4)$$

Since the starting term in the first sum is 0, effectively both the sides are sums of 100 terms each and it would be great if they matched term by term. But this is too optimistic. Even for $j = 1$, $jb_j = b_1$, the number of students having exactly one friend which will generally be less than c_1 , the number of students having *at least* one friend.

To prove (4), we need to express each b_j in terms of some c 's. This can be done as follows. Denote the set of all 100 students by S . For every j , $0 \leq j \leq 99$, let S_j be the subset of those students who have at least j friends. Clearly S_0 is the entire set S . As j increases, the sets S_j go on shrinking so that we have a descending chain of subsets of S

$$S = S_0 \supset S_1 \supset S_2 \dots \supset S_j \supset S_{j+1} \supset \dots S_{99} \quad (5)$$

For every j , note that

$$T_j = S_j - S_{j+1} \quad (6)$$

because if a student has exactly j friends, that puts him in S_j but outside S_{j+1} . Taking cardinalities (i.e. the numbers of elements) of both the sides, we get

$$b_j = c_j - c_{j+1} \quad (7)$$

for $j = 0, 1, \dots, 99$ (We set $c_{100} = 0$, since no student can have more than 99 friends.) There is little point in forming the sum $\sum_{j=0}^{99} b_j$ because in view of (7) this will be a telescopic series whose sum will be $c_0 - c_{100}$, i.e. 100, which we know already as there are 100 students.

But let us consider the sum $\sum_{j=1}^{99} jb_j$. Again because of (7) we have

$$\sum_{j=1}^{99} jb_j = \sum_{j=1}^{99} jc_j - \sum_{j=1}^{99} jc_{j+1} \quad (8)$$

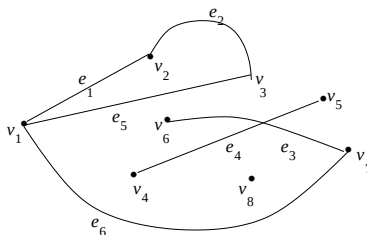
Note that c_j has coefficient j in the first sum on the R.H.S. but it has coefficient $j - 1$ in the second sum. So, we can rewrite the second sum

as $\sum_{j=1}^{99} (j-1)c_j$. (We have to account for the missing term $99c_{100}$. But that is 0.) So (8) gives

$$\sum_{j=1}^{99} j b_j = \sum_{j=1}^{99} (j - (j-1))c_j = \sum_{j=1}^{99} c_j \quad (9)$$

which is the same as (3) and the solution is complete. ■

The reason we denoted the students by $v_1, v_2, \dots, v_{100}$ is that we can give the proofs of (1) and (2) in a graphic way. A **graph** consists of some points (called vertices) and some edges joining them. The edges need not be straight line segments. Also when you draw a graph in a plane, it is quite all right if the edges intersect at some points other than vertices. We simply ignore such crossings. The only restrictions are that there be at most one edge joining two distinct points and that the two points joined by an edge be distinct. (It is possible to relax these restrictions to get what is called a **multigraph**. But we shall not go into it.) The figure below shows a graph with 8 vertices v_1 to v_8 and six edges labelled e_1 to e_6 .



Our problem could be represented by a graph with 100 vertices labelled v_1 to v_{100} in which v_i and v_j are joined by an edge if and only if they are friends of each other. The number a_i of friends which v_i has is the number of edges that meet at v_i and is called the **degree** (or the **valency**) of the vertex v_i . In the figure above, the degrees of the vertices v_1 to v_8 are 3, 2, 2, 1, 1, 1, 2 and 0 respectively. If we add them we get 12, which is twice the number of edges. This holds in every graph and proved by the same reasoning as (1) above. Let us also classify the 8 vertices according to their degrees. Then we have

$b_0 = 1, b_1 = 3, b_2 = 3, b_3 = 1$ and $b_j = 0$ for all $j > 3$. If we form the sum $\sum_{j=0}^7 j b_j$, it is $3 + 6 + 3 = 12$ again. This is true in any graph and generalises (2).

The first part of the solution was to obtain (3). The technique used here is called **double counting**. It consists of counting the number of elements in some set or some related number (e.g. $2|F|$ in our problem) in two different ways and equating the two counts. A familiar example of double counting is the identity $\sum_{k=0}^n \binom{n}{k} = 2^n$. Here the k -th term on the L.H.S. is the number of k -subsets (i.e. subsets having k elements) of an n -set, i.e. a set, say S with n elements. So the L.H.S. is the total number of subsets of S . However, every such subset, say A of S determines a unique function $f : S \rightarrow \{0, 1\}$ defined by $f(x) = 1$ if $x \in A$ and $f(x) = 0$ if $x \notin A$. As the number of all functions from S to $\{0, 1\}$ is 2^n , the identity is proved.

The second part of the solution is more tricky. In the problem, we are given c_j to be the number of students having at least j friends. If, instead, c_j were the number of students having *at most* j friends, then the sequence $\{c_j\}$ will be increasing rather than decreasing as j increases from 0 to 99, with $c_{99} = 100$. b_j would now equal $c_j - c_{j-1}$. If we replace b_j by $c_j - c_{j-1}$, then the equality (3) we get by double counting becomes

$$\sum_{i=1}^{100} a_i = \sum_{j=0}^{99} j(c_j - c_{j-1}) \quad (10)$$

The sum on the R.H.S. is the discrete analogue of what is called a **Riemann-Stielje integral** of the function $f(j) = j$ w.r.t. the measure c . This topic is far too advanced to permit any further discussion here. But it is important in statistics. Maybe this problem was designed by a statistician, a highly probable event considering that the test is for admission to a reputed institute of statistics!

In the version of the question paper on the ISI website, in the statement of the question, the second sum starts with $j = 0$ instead of $j = 1$. That seems to be a misprint.

- Q. 2.** It is given that the graph of $y = x^4 + ax^3 + bx^2 + cx + d$ (where a, b, c, d are real) has at least 3 points of intersection with the x -axis. Prove that either there are exactly 4 distinct points of intersection, or one of those 3 points of intersection is a local minimum or maximum.

Solution and Comments: Let $f(x) = x^4 + ax^3 + bx^2 + cx + d$. Being a polynomial of degree 4 it has 4 roots, some of which may be complex. But as the degree is even and all coefficients are real, the number of real roots is even. We are given that it already has at least three distinct roots, say α, β, γ . So the fourth root, say δ , must also be real. If δ is different from all these roots, then $f(x)$ has exactly four distinct roots. If $\delta = \alpha$ (say), then $f(x)$ factors as $(x - \alpha)^2 g(x)$, where

$$g(x) = (x - \beta)(x - \gamma) \quad (1)$$

Direct differentiation gives

$$f'(x) = 2(x - \alpha)g(x) + (x - \alpha)^2 g'(x) \quad (2)$$

$$\text{and } f''(x) = 2g(x) + 2(x - \alpha)g'(x) + (x - \alpha)^2 g''(x) \quad (3)$$

From (2), $f'(\alpha) = 0$. But from (3), $f''(\alpha) = 2g(\alpha) = 2(\alpha - \beta)(\alpha - \gamma) \neq 0$ since $\alpha \neq \beta$ and $\alpha \neq \gamma$. So f has a local minimum or a local maximum at $x = \alpha$ according as $g(\alpha) < 0$ or $g(\alpha) > 0$. ■

An extremely simple and hence disappointing problem. Probably to make up for its simplicity, instead of saying that $f(x)$ has at least three distinct real roots, the data is given in a slightly twisted form, viz. that its graph cuts the x -axis at at least three points. Any candidate who knows that a (sufficiently differentiable) function has a local extremum at a point where it has a zero of order 2 would be able to answer the question instantaneously. Probably the idea is to see if the candidate can present the reasoning correctly, a fast declining quality these days.

- Q. 3.** Consider a triangle PQR in $\mathbb{R}^2$. Let A be a point lying on PQR or in the region enclosed by it. Prove that, for any function $f(x, y) = ax + by + c$ on $\mathbb{R}^2$,

$$f(A) \leq \max\{f(P), f(Q), f(R)\}.$$

Solution and Comments: Note that the coordinates of the vertices of the triangle are not given. In such problems, it pays to paraphrase the problem so that it becomes easier to consider some analogous but simpler problem and see if its solution provides any clue to solve the given problem.

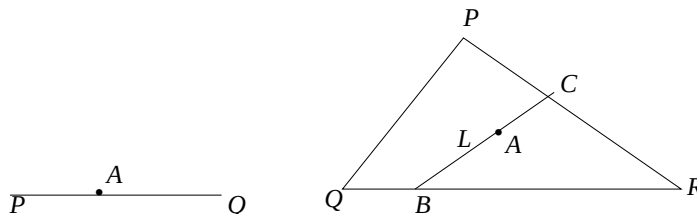
In the present problem, such a paraphrase is that the maximum of $f(x, y)$ occurs when (x, y) is some vertex of the triangle. A triangle is an object in $\mathbb{R}^2$ having three vertices. Its prototype is an interval, or a segment, say PQ , in $\mathbb{R}$. A point of $\mathbb{R}$ is specified by only one number, say x . The appropriate one dimensional prototype of the function $f(x, y) = ax + by + c$ would be a function of the form $f(x) = ax + b$ where a, b are some constants. So the one dimensional analogue of the result we have to prove is that if A is any point of an interval with end points P and Q (as shown in the first part of the figure below), then

$$f(A) \leq \max\{f(P), f(Q)\} \quad (1)$$

where f is a function of the form

$$f(x) = ax + b \quad (2)$$

Or put differently, the function given by (2) attains its maximum on the segment at one of the end-points.



This is very easy except for those who are thoroughly brainwashed by calculus to think that derivatives are the only tool to find maxima and minima. Here calculus fails miserably because from (2)

$$f'(x) = a \quad (3)$$

which never vanishes unless $a = 0$. (If $a = 0$, $f(x)$ is a constant function b .) So the calculus method is nipped in the bud. But with or without

calculus, $f(x)$ is an increasing function of x if $a > 0$ and a decreasing function if $a < 0$. Therefore, if P, Q correspond to the real numbers p, q (with $p < q$), then on the closed interval $f(x)$ attains its maximum at Q while if $a < 0$, then $f(x)$ attains its maximum at P . So in either case, (1) is true. (In the degenerate case where $a = 0$, f attains both its maximum and minimum at every point of PQ . But that does not render the statement false, because to say that the maximum is attained at a point P does not mean that it is attained *only* at P .)

How does this help in proving (2)? We now have a function

$$f(x, y) = ax + by + c \quad (4)$$

of two variables and the problem is to maximise it over a two dimensional domain, viz. the triangle PQR and the region enclosed by it? (It is customary to call this domain as the **closed**¹ **triangle** PQR . There is a calculus based method for finding the maxima and the minima of functions of two variables. But it is beyond the HSC level. That does not matter here because even for a function (2) of one variable x , calculus was not of much help.

To prove (4), we proceed in two stages. First we show that the maximum must occur at some boundary point of the triangle, i.e. at some point on one of its three sides. This amounts to showing that f cannot attain its maximum at an interior point A such as the one shown in the second part of the figure. Given any such point we draw any line L , with an equation, say $y = mx + u$, through it which cuts two of the sides at points B and C as shown. A typical point on the segment BC will be of the form $(x, mx + u)$ where x varies from the x -coordinate of B to that of C . (In essence, we are parametrising the segment BC with x as a parameter.) For points on this segment we have

$$f(x, y) = f(x, mx + u) = ax + b(mx + u) + c = (a + mb)x + (bu + c) \quad (5)$$

which is very similar to (2) except that the coefficients and the constant terms are different. So by the same logic that we used to maximise the

¹More generally, the term ‘closed’ is used to describe a set which includes all its boundary points. This is in analogy with closed intervals. Thus a closed disc means a disc including the boundary circle.

function (2), f will attain its maximum on this segment at either B or C . It is not important to identify the line L or the points B and C . What matters is that there is at least one boundary point at which f has a higher value than at A . (Again, in the degenerate case when both a, b vanish in (4), it becomes a constant function and then the result asked holds trivially.)

Thus the search for the maximum is narrowed down to the boundary of the triangle PQR , which consists of three segments PQ, QR and RP . It is now easy to finish it. We parametrise each of these segments and by the same reasoning above, see that on each one of them f is maximum at one of its two end points. All points of the triangle are now exhausted and thus we have proved that the search ends with one of the vertices P, Q, R as the location for the maximum of f . (It is also true that the minimum of f occurs at one of the three vertices. The proof is similar. Or, following a common trick, we apply the earlier part to $-f$ and maximise it.) ■

A function such as (2) is called **linear** for the simple reason that its graph is a straight line. For a function like (4) the graph is a subset of $\mathbb{R}^3$, viz. $\{(x, y, z) : z = ax + by + c\}$. It is a plane and not a straight line. But the term ‘linear’ is used for such functions too. What we have proved here is that a linear function attains its maximum on a closed triangle at some vertex of it. A similar result holds for the minimum of a linear function over a closed triangle. In fact, we can replace the triangle by any (closed) polygon. More generally, a function of n real variables of the form

$$f(x_1, x_2, \dots, x_n) = c_1x_1 + c_2x_2 + \dots + c_nx_n + \alpha \quad (6)$$

where $c_1, c_2, \dots, c_n$ and α are some constants is said to be a linear function. (The constant α has no role in maximisation and can as well be dropped.) The concept of a polygon can also be generalised in higher dimensions to polyhedra. (A tetrahedron and a parallelepiped are examples of such polyhedra in $\mathbb{R}^3$. In higher dimensions, the word used sometimes is ‘hyper-polyhedron’.)

Essentially the same reasoning as that in the solution of the present problem shows that a linear function attains its extrema on a closed polyhedron at its vertices. This result is the basis of an important and

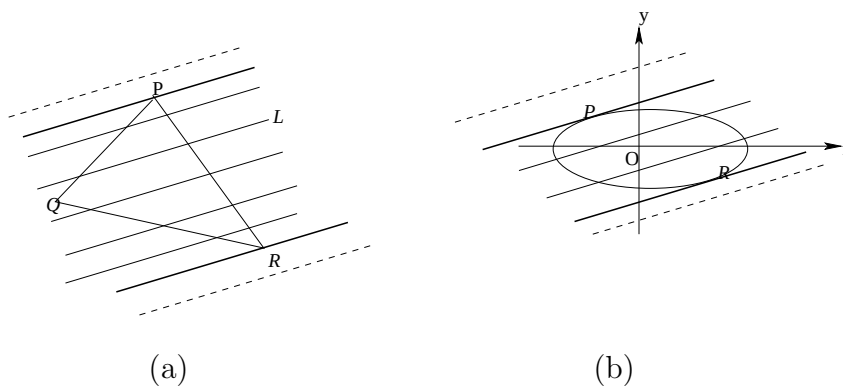
highly applicable branch of mathematics, called **linear programming**. There is no such simple result for non-linear functions. For example, the function $f(x, y) = x^2 + y^2 + 10$ attains its minimum (viz. 10) at the origin which may very well be an interior point of a triangle.

We now give another solution which is based on the concept of what are called **level curves** of a function $f(x, y)$ of two variables. These are curves of the form

$$\{(x, y) \in \mathbb{R}^2 : f(x, y) = c\} \quad (7)$$

where c is a constant. As the constant changes, we get a different level curve of the same function. The name ‘level curve’ comes from the fact that if we consider the graph of the function, it is the surface $S = \{(x, y, z) \in \mathbb{R}^3 : z = f(x, y)\}$ and the points on it above the curve (7) all lie on a horizontal plane $z = c$. (The corresponding curve on S is called a **contour line** and sometimes this term is also used for level curves.) Depending on the nature of the function $f(x, y)$ its level curves are given special names. For example, in a weather map, isotherms are the level curves of the temperature function while isobars are the level curves of the barometric pressure. All the conics and many other plane curves can be expressed as level curves of suitable functions.

The different level curves of a function often have some common feature. For example, those of a linear function are a family of parallel straight lines while those of the function $f(x, y) = x^2 + y^2$ are concentric circles. Our alternate solution is based on the former fact.



The level curves of the function $f(x, y) = ax + by + c$ will be straight lines of the form $\{(x, y) : ax + by + c = r\}$. For some values of r this

line will intersect the closed triangle PQR . One such line L is shown in (a) of the figure above. As r increases, the line moves parallel to itself in one direction and as r decreases it moves in the opposite direction. When r is either too large or too small the corresponding level curves (shown by dotted lines) do not intersect the triangle.

The key idea is that if f attains its maximum on the triangle PQR at a point A , then the triangle must lie entirely on one side of the level curve passing through A for otherwise the function will have a higher value at points of the triangle on one side of the level curve. Therefore A must be a vertex of the triangle. (If the function is such that its level curves happen to be parallel to one of the sides of the triangle, then all the points of this side will be points of maximum or minimum.) Put differently, the extrema of the function f occur just at the points where the moving family of lines parallel to L ‘exit’ the triangle PQR . These departing lines are shown in (a) by thickening.

In fact this reasoning holds true even if the triangle PQR is replaced by some other closed set, say the ellipse and the region bounded by it as shown in (b). Then a linear function will attain its maximum as well as its minimum on this set at points on the ellipse where the level curves touch it. So, even without knowing the theory of maxima/minima of functions of two variables, in the special case where the function is linear, these points can be located by finding points on the boundary where the tangent is parallel to the level curves. (This breaks down for a triangle because the boundary of a closed triangle is not a smooth curve. There is no tangent at a vertex.)

- Q. 4.** Let f and g be two non-decreasing twice differentiable functions defined on an interval (a, b) such that for each $x \in (a, b)$, $f''(x) = g(x)$ and $g''(x) = f(x)$. Suppose also that $f(x)g(x)$ is linear in x on (a, b) . Show that we must have $f(x) = g(x) = 0$ for all $x \in (a, b)$.

Solution and Comments: The hypothesis about linearity in x means that there exist some constants u, v such that

$$f(x)g(x) = ux + v \quad (1)$$

for all $x \in (a, b)$. (The concept of a linear function of two variables also came up in the data of the last question. But that time these words

not used and instead a formula for the function was given. This time, we are only given the name ‘linear’ and expected to figure out what it means. Such a duplicity, especially in two consecutive questions in the same test, is unbecoming to say the least. If the candidates are expected to know what linearity of a function means, better to use that phrase in both the questions. Otherwise give a formula in both the questions. Adopting one approach in one question and another in the second can be confusing to a candidate by making him wonder whether something else is meant which he is missing.)

Anyway, coming to the question, differentiating both the sides of (1) twice, we get

$$f''(x)g(x) + 2f'(x)g'(x) + f(x)g''(x) = 0 \quad (2)$$

As we are given $f''(x) = g(x)$ and $g''(x) = f(x)$ for all $x \in (a, b)$, this becomes

$$(g(x))^2 + 2f'(x)g'(x) + (f(x))^2 = 0 \quad (3)$$

for all $x \in (a, b)$. The first and the last terms are always non-negative and so is the middle one because $f(x)$ and $g(x)$ are given to be differentiable and non-decreasing which means that $f'(x), g'(x)$ are both non-negative everywhere.

So the net effect is that the sum of three non-negative terms is 0. This forces each one of them to vanish. So, in particular, $f(x) = g(x) = 0$ for all $x \in (a, b)$. ■

Apparently, the only purpose in asking this question is to test the knowledge of the fact that linear functions of one variable have identically vanishing second derivatives. (The converse is also true but not needed in the solution.) Perhaps that is why the term ‘linear’ was used. If instead, an equation like (1) were given as hypothesis, it would be too obvious to begin by differentiating it twice. Anyway, once this idea strikes there is little else in the problem. Actually, a condition like (1) is very strong because it means that $g(x)$ will equal $\frac{ux+v}{f(x)}$ and then it is highly unlikely that $g''(x)$ would equal $f(x)$ except in a degenerate case, which is the assertion of the problem. Otherwise

$f(x)$ will satisfy a fairly complicated second order differential equation, whose solution can hardly be expected at this level.

In fact, using differential equations of higher orders, we can give a more sophisticated solution. Although f and g are given to be only twice differentiable, from the strong conditions $f''(x) = g(x)$ and $g''(x) = f(x)$ we see that they have derivatives of all orders. Also the fourth derivative $f^{(4)}(x)$ equals $g''(x)$ which in turn equals $f(x)$. The same holds for the fourth derivative of $g(x)$. In other words both $f(x)$ and $g(x)$ are solutions of the fourth order differential equation

$$\frac{d^4 y}{dx^4} - y = 0 \quad (4)$$

Although the general theory of linear differential equations with constant coefficients is beyond the HSC level, because of applications like simple harmonic motion, many students are familiar with the solutions of second order linear differential equations. Assuming such familiarity, we note that the characteristic roots of the equation above are ± 1 and $\pm i$. So, in a real form the general solution is

$$y = c_1 e^x + c_2 e^{-x} + c_3 \cos x + c_4 \sin x \quad (5)$$

where c_1, c_2, c_3, c_4 are arbitrary constants. To determine a particular solution, we need four conditions about the solution. In the present problem no such conditions about $f(x)$ or $g(x)$ are given. So all we can say is that

$$f(x) = a_1 e^x + a_2 e^{-x} + a_3 \cos x + a_4 \sin x \quad (6)$$

$$\text{and } g(x) = b_1 e^x + b_2 e^{-x} + b_3 \cos x + b_4 \sin x \quad (7)$$

where the a 's and the b 's are some constants. Although we have no way of determining them, no matter what they are, the product $f(x)g(x)$ will equal a linear combination of many functions such as $e^{2x}, 1, e^{-x} \cos x$ etc. But the function x is not in this list. So if (1) is to hold then u must vanish and so we get

$$f(x)g(x) = v \quad (8)$$

where v is some constant. From (6) and (7), this can happen in only two cases, either (i) $f(x) = a_1 e^x$, $g(x) = b_2 e^{-x}$ with $a_1 b_2 = v$ or (ii)

$f(x) = a_2 e^{-x}$, $g(x) = b_1 e^x$ with $a_2 b_1 = v$. Without loss of generality, we assume that (i) holds. But then $f''(x) = a_1 e^x$ which cannot equal $g(x)$ unless both a_1 and b_2 both vanish. But in that case both $f(x)$ and $g(x)$ are identically 0. ■

Although the second solution involves ideas slightly beyond the HSC level, it is less tricky than the first solution. The first solution crucially requires that $f'(x), g'(x)$ are non-negative and if it does not strike you to use this then from (3) you cannot conclude anything.

- Q. 5.** Show that the sum of 12 consecutive integers can never be a perfect square. Give an example of 11 consecutive integers whose sum is a perfect square.

Solution and Comments: Suppose S is the sum of 12 consecutive integers. Call the middle two integers as n and $n + 1$. Then $S = 6(2n + 1)$ which can never be a perfect square because it is divisible by 2 but not by 4.

Suppose, however, that S is the sum of 11 consecutive integers. This time let n be the (unique) middle term. Then $S = 11n$ which will be a perfect square whenever n is 11 times some perfect square, e.g. $n = 11, 44$. ■

A surprisingly simple problem. The success lies in the choice of the notations. We could have taken the integers starting from some m . But then the expression for S would be complicated. The problem is a good test of the ability to choose convenient notations.

- Q. 6.** Let A be the region in the xy -plane given by

$$A = \{(x, y) : x = u + v, y = v, u^2 + v^2 \leq 1\}.$$

Derive the length of the longest line segment that can be enclosed inside the region A .

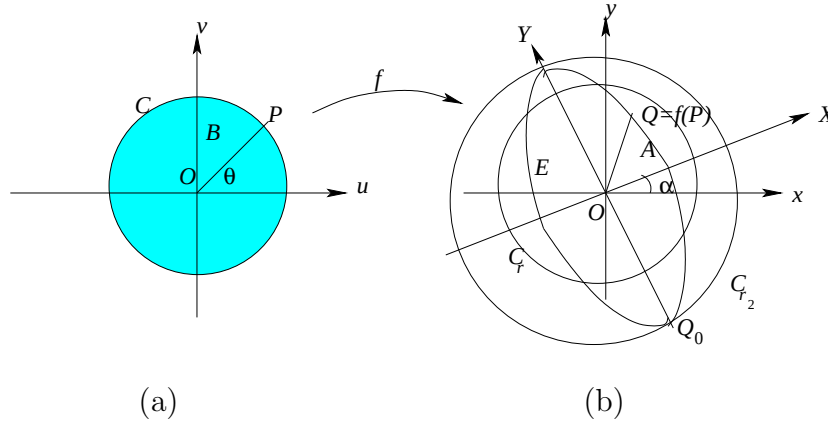
Solution and Comments: Normally, a region in the xy -plane is specified by one or more inequalities satisfied by some functions of the coordinates x, y . For example, the single inequality $x^2 + y^2 \leq 1$

gives the closed unit disc D centred at the origin. But $D^+ = \{(x, y) : x^2 + y^2 \leq 1, y \geq 0\}$ is the upper half of this disc. The region determined by three inequalities, say $\{(x, y) : x \geq 0, y \geq 0, x + y \leq 1\}$ is a (closed) triangle with vertices at $O, (1, 0)$ and $(0, 1)$. Generally the boundary of such a region consists of several pieces obtained by converting one inequality at a time to an equality. For example, the boundary of D^+ consists of the union of the semi-circular arc $x^2 + y^2 = 1, y \geq 0$ and the set $\{(x, y) : y=0, x^2 \leq 1\}$ which is the diameter through O .

In the present problem, however, the region A is not specified directly in terms of any inequalities involving x and y . Instead, x and y are given as functions of some other variables u and v and an inequality satisfied by u and v , viz.

$$u^2 + v^2 \leq 1 \quad (1)$$

is given. In the uv -plane the region, say B , defined by this inequality is the unit disc shown in (a) of the figure below.



We now consider a function f from the uv -plane to the xy -plane defined by $f(u, v) = (x, y)$ where

$$x = u + v \quad (2)$$

$$\text{and } y = v \quad (3)$$

So, the region A in which we are interested is the image of the region B under this function f . We can express it directly in terms of an

inequality in x, y if we eliminate u, v from (1), (2) and (3). Putting $v = y$ and $u = x - v = x - y$ into (1) and simplifying, we get

$$A = \{(x, y) : x^2 - 2xy + 2y^2 \leq 1\} \quad (4)$$

The boundary of A will consist of those points where equality holds. So, it will be the curve, say E , whose equation is

$$x^2 - 2xy + 2y^2 = 1 \quad (5)$$

This is of the form

$$Ax^2 + 2Hxy + By^2 = 1 \quad (6)$$

where $A = 1, H = -1$ and $B = 2$. Since $H^2 < AB$, E is an ellipse. But it is not in the standard form. If we change the coordinate axes to OX and OY which are inclined at an angle α with the old axes as shown in (b), then

$$x = X \cos \alpha - Y \sin \alpha \quad (7)$$

$$\text{and } y = X \sin \alpha + Y \cos \alpha \quad (8)$$

If we substitute these into (5) we shall get an equation of the form

$$A'X^2 + 2H'XY + B'Y^2 = 1 \quad (9)$$

where A', B' and H' are expressed in terms of A, B, H and α . If α is chosen so that H' vanishes then in the new coordinate frame, the equation of E will be

$$\frac{X^2}{a^2} + \frac{Y^2}{b^2} = 1 \quad (10)$$

where $a = \frac{1}{\sqrt{A'}}$ and $b = \frac{1}{\sqrt{B'}}$. This is in the standard form and we can then answer any questions about the region such as its area, the eccentricity of its boundary E and so on. This is a standard, but rather elaborate method, the details of which are given in Comment No. 18 of Chapter 10. But we do not follow it because our interest is merely to find the length of the longest line segment that can be enclosed inside the region A . It is *not* required to identify these two farthest points.

With a modest goal, there is often a short cut. And, that is indeed the case here. Knowing that the region is bounded by an ellipse E , this farthest pair must be the extremities of the major axis of the ellipse. Also each of these extremities is the farthest point on the ellipse from its centre which is the origin O .

Thus our problem is reduced to finding the maximum distance of a point on the ellipse E given by (5) from the origin. A typical point, say Q , on E will be of the form $f(P)$ for some point P on the circle C which is the boundary of the region B in the uv -plane. Taking the standard parametrisation of C as

$$u = \cos \theta, v = \sin \theta, \quad 0 \leq \theta \leq 2\pi \quad (11)$$

the problem is reduced to maximising the function

$$\begin{aligned} f(\theta) &= x^2 + y^2 = (u + v)^2 + v^2 \\ &= (\cos \theta + \sin \theta)^2 + \sin^2 \theta \\ &= 1 + \sin^2 \theta + 2 \sin \theta \cos \theta \\ &= \frac{3}{2} + \sin 2\theta - \frac{1}{2} \cos 2\theta \end{aligned} \quad (12)$$

To maximise $f(\theta)$ we can begin by finding $f'(\theta)$ and setting it equal to 0. But there is a short cut here too. The first term on the R.H.S. of (12) is a constant. We apply the Cauchy Schwarz inequality to the sum of the other two terms to get

$$|\sin 2\theta - \frac{1}{2} \cos 2\theta| \leq \sqrt{1 + \frac{1}{4}} \sqrt{\sin^2 2\theta + \cos^2 2\theta} = \frac{\sqrt{5}}{2} \quad (13)$$

So the square of the distance of either end of the major axis of E from the centre is $\frac{3 + \sqrt{5}}{2}$. Hence the major axis has length $\sqrt{2} \sqrt{3 + \sqrt{5}}$ and this is the length of the longest segment that can be enclosed in the given region A . ■

Note that equality will hold in (13) when $\sin 2\theta$ and $\cos 2\theta$ are proportional to 1 and $-\frac{1}{2}$, i.e. when

$$\tan 2\theta = -2 \quad (14)$$

There are four solutions of this equation in the interval $[0, 2\pi]$. Two of them will give the points on E that are closest to O while the other two will give points that are farthest. So, even if we were asked to identify the points in A which are farthest apart, and not just how far they are apart from each other, this method will work.

The method of level curves which we introduced in the alternate solution to Q.3 above can also be applied here to find the farthest points on the ellipse from the origin. In essence we want to maximise the function

$$g(x, y) = x^2 + y^2 \quad (15)$$

on the ellipse E . (Actually, we want to maximise it on the closed region A bounded by the ellipse. But it is obvious that the maximum will occur only at a boundary point, because given any interior point R , if we let the ray (also called the half line) from O through R cut E at S , then S will be farther from O than R is. So the search for the farthest point is confined to the boundary E .)

The level curves of $g(x, y)$ are concentric circles centred at O . One such circle C_r having radius r is shown in (b) of the figure. Now let r_1 and r_2 be the minimum and the maximum distances of points on E from O . (Since E is known to be an ellipse with centre O , r_1 and r_2 are nothing but the semi-minor and the semi-major axis of the ellipse. But we don't know them right now. Rather our goal is to find r_2 .) It is obvious that for $r < r_1$, C_r will lie entirely inside E while for $r > r_2$, E will lie entirely inside C_r . In both these cases, C_r will not intersect E . But if $r_1 < r < r_2$, then C_r will meet the ellipse E at four points. For $r = r_1$, C_r will touch E internally while for $r = r_2$, C_r will touch E externally. Our interest is in finding the latter. (C_{r_2} is also shown in the figure.)

To find C_{r_2} , we let $Q_0 = (x_0, y_0)$ be its point of contact with E . (Actually, there will be two points of contact. But the other one will be $(-x_0, -y_0)$. So it suffices to find either one of them.) Then the tangent to C_{r_2} at Q_0 has the same slope as the tangent to E at Q_0 . The former is $-\frac{x_0}{y_0}$. The latter can be found by writing down the equation of the tangent to (5) at (x_0, y_0) , viz.

$$xx_0 - xy_0 - yx_0 + 2yy_0 = 1 \quad (16)$$

which has slope $\frac{y_0 - x_0}{2y_0 - x_0}$. (One can also get this by implicit differentiation. But in principle, that is the same method.) Equating the two slopes,

$$\frac{y_0 - x_0}{2y_0 - x_0} + \frac{x_0}{y_0} = 0 \quad (17)$$

We have to solve this simultaneously with

$$x_0^2 - 2x_0y_0 + 2y_0^2 = 1 \quad (18)$$

The system of these two equations will have four solutions. Two of them will give the ends of the major axis and the other two the ends of the minor axis of E . Being a non-linear system, there is no golden method for solving it. But we note that on simplification, (17) becomes

$$y_0^2 + x_0y_0 - x_0^2 = 0 \quad (19)$$

Substituting $x_0y_0 = x_0^2 - y_0^2$ into (18) we get

$$-x_0^2 + 4y_0^2 = 1 \quad (20)$$

If we put $y_0 = mx_0$ into (19), we get a quadratic in m . viz.

$$m^2 + m - 1 = 0 \quad (21)$$

The same substitution in (20) gives

$$x_0^2 = \frac{1}{4m^2 - 1} = \frac{1}{3 - 4m} \quad (22)$$

and hence

$$y_0^2 = m^2x_0^2 = \frac{m^2}{3 - 4m} = \frac{m - 1}{4m - 3} \quad (23)$$

Thus we would know x_0 and y_0 if we know the value of m . Our interest, however, is more in $x_0^2 + y_0^2$ which we get by adding (22) and (23).

$$x_0^2 + y_0^2 = \frac{m - 2}{4m - 3} \quad (24)$$

It only remains to find the value of m . From (21) there are two possible values of m , viz. $\frac{-1 \pm \sqrt{5}}{2}$. One value will give r_1^2 and the other r_2^2 . As we already know r_2 as $\frac{3 + \sqrt{5}}{2}$ by another method we leave the calculation to the reader.

In passing we also remark that although earlier we abandoned the conversion of (5) to (9) as complicated, for those who are familiar with such reduction to the standard form, there are ready made formulas for expressing A' , B' and H' in terms of A , B , H and α . They are given at the end of p. 363. In order to make H' vanish, α has to be chosen as $\frac{1}{2} \tan^{-1} \frac{2H}{A-B}$. With this choice of α , the formulas for A' and B' simplify to

$$A' = \frac{1}{2}[A + B + \sqrt{4H^2 + (A - B)^2}] \quad (25)$$

$$\text{and } B' = \frac{1}{2}[A + B - \sqrt{4H^2 + (A - B)^2}] \quad (26)$$

In the present problem, $A = 1$, $B = 2$ and $H = -1$. So we have $A' = \frac{3 + \sqrt{5}}{2}$ and $B' = \frac{3 - \sqrt{5}}{2}$. Therefore in (10) we now have

$$a = \sqrt{\frac{2}{3 + \sqrt{5}}} \quad (27)$$

$$\text{and } b = \sqrt{\frac{2}{3 - \sqrt{5}}} \quad (28)$$

Note that $b > a$ and so the major axis of E lies along OY and not along OX . So the distance between the farthest points on the ellipse is $2b = 2\sqrt{\frac{2}{3 - \sqrt{5}}}$. It is not immediately obvious that this is the same answer as $\sqrt{2} \sqrt{3 + \sqrt{5}}$ which we got earlier by a different method. To check that they are indeed the same we show that their squares are equal, which boils down to showing that $\frac{8}{3 - \sqrt{5}} = 2(3 + \sqrt{5})$. But that is true because $\frac{1}{3 - \sqrt{5}} = \frac{3 + \sqrt{5}}{9 - 5} = \frac{3 + \sqrt{5}}{4}$. Note also that the

angle α through which we rotated the axes is given by

$$\tan 2\alpha = \frac{2H}{A - B} = 2 \quad (29)$$

For the angle θ in (14), we had $\tan 2\theta = -2$. But α and θ are not related because θ was the angle the line OP made with the u -axis in the uv plane and there is no reason to suppose that OQ (where $Q = f(P)$) makes the same angle with the x -axis in the xy -plane.

The problem is very well designed and a reminder of the golden days of JEE. The first part, that of specifying the region A as the image of some other region under a function is especially novel. The second part is comparatively routine. But even here a candidate who has the ability to see the short cuts gets duly rewarded over a candidate who spends time in bringing the ellipse to a standard form. Of course those who have very good memory to write (25) and (26) have an advantage. But since the problem gives the region parametrically (in terms of some functions of the parameters u and v) a solution based on these parameters is simpler. If the problem was given directly as to find the farthest points in the region bounded by the curve (5), then the second approach is better. Even then, one can go backwards and write $x^2 - 2xy + 2y^2$ as $(x - y)^2 + y^2$ and then put $u = x - y, v = y$. That will give the present problem.

Q. 7. Let $f : [0, \infty) \rightarrow \mathbb{R}$ be a non-decreasing continuous function. Show then that the inequality

$$(z - x) \int_y^z f(u) du \geq (z - y) \int_x^z f(u) du$$

holds for any $0 \leq x < y < z$.

Solution and Comments: As both $z - x$ and $z - y$ are positive, we can rewrite the inequality to be proved as

$$\frac{1}{z - y} \int_y^z f(u) du \geq \frac{1}{z - x} \int_x^z f(u) du \quad (1)$$

In this form, the L.H.S. is the average value of the function $f(u)$ over the interval $[y, z]$ while the R.H.S. is the average value of this function

over the bigger interval $[x, z]$. Now the inequality is intuitively obvious. Think of u as time and $f(u)$ as the profit at time u . We are given that the profit function is non-decreasing. So the later portions of the time interval $[x, z]$ are more prosperous than the earlier ones. So it is obvious that the average taken over a period of greater prosperity is more than the average over the entire period which also includes some lean time initially.

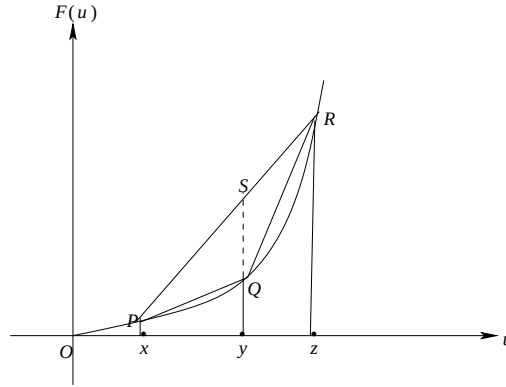
To give a rigorous proof, we apply the Fundamental theorem of calculus (FTC) and the Lagrange Mean Value Theorem (LMVT). Let

$$F(u) = \int_x^u f(t)dt \quad (2)$$

for $u \geq 0$. The inequality (1) to be proved now takes the form

$$\frac{F(z) - F(y)}{z - y} \geq \frac{F(z) - F(x)}{z - x} \quad (3)$$

which can be interpreted geometrically in terms of the slopes of certain chords joining some points on the graph of $F(u)$. This graph is shown in the figure below.



Let $P = (x, F(x))$, $Q = (y, F(y))$ and $R = (z, F(z))$ be the points on this graph as shown. Then (2) amounts to showing that

$$m(QR) \geq m(PR) \quad (4)$$

where m stands for the slope. Actually, a double inequality holds, viz.

$$m(QR) \geq m(PR) \geq m(PQ) \quad (5)$$

Verbally, the slope of the chord PR lies between the slope of the chord PQ and that of the chord QR . Again this is very obvious intuitively in the present case because these slopes are the average profits over the respective time intervals and common sense tells us that the average profit over an entire interval will lie somewhere between the average profit over a leaner portion of the interval and that over a prosperous portion.

For a rigorous proof we need that the function $F(u)$ is concave upwards on $[0, \infty)$. This means that the chord lies above the curve. In other words, if the line $u = y$ meets the chord PR at a point S , then S lies above Q . Once this is accepted the proof of (5) is immediate because the middle term $m(PR)$ equals $m(PS)$ and also $m(SR)$.

To show that $F(u)$ is concave upwards, we use FTC. As f is given to be continuous, by the FTC we have

$$F'(u) = f(u) \tag{6}$$

for all $u \geq 0$. But we are also given that $f(u)$ is non-decreasing. So $F'(u)$ is a non-decreasing function of u and this is a sufficient condition for $F(u)$ to be concave upwards on the interval $[0, \infty)$. (This is a standard application of LMVT. The condition for concavity upwards is often stated in terms of the second derivative, viz. $F''(u) \geq 0$ for all $u \in (0, \infty)$. But the proof given in Comment No. 11 of Chapter 16) goes through if we merely know that $F'(u)$ is non-decreasing as in the present case.) ■

As a matter of fact, we do not need the FTC and the LMVT. We can give a direct proof based on what is called the **Mean Value Theorem for Integrals**. It says that the average value of a continuous function over an interval is attained at some point of that interval. (Although this is a non-trivial result, its proof does not require the FTC. It is an easy consequence of the Intermediate Value Property for continuous functions.) We apply this theorem separately to the intervals $[x, y]$ and $[y, z]$. Call the integrals $\int_x^z f(u)du$, $\int_x^y f(u)du$ and $\int_y^z f(u)du$ as I , I_1 and I_2 respectively. Then by the additivity property of the integrals,

$$I = \int_x^z f(u)du = \int_x^y f(u)du + \int_y^z f(u)du = I_1 + I_2 \tag{7}$$

Applying the MVT for integrals to I_1 and I_2 we get that

$$I_1 = (y - x)f(c_1) \quad (8)$$

$$\text{and } I_2 = (z - y)f(c_2) \quad (9)$$

for some $c_1 \in [x, y]$ and some $c_2 \in [y, z]$. Obviously, $c_1 \leq c_2$ and as f is non-decreasing,

$$f(c_1) \leq f(c_2) \quad (10)$$

an inequality which we shall use crucially. But first let us express $y - x$ and $z - y$ as some fractions of $z - x$. Let

$$\lambda = \frac{y - x}{z - x} \quad (11)$$

Since $x < y < z$ we have $\lambda \in (0, 1)$. Further by a direct computation,

$$1 - \lambda = \frac{z - y}{z - x} \quad (12)$$

Substituting these into (8) and (9) we get

$$I_1 = \lambda(z - x)f(c_1) \quad (13)$$

$$\text{and } I_2 = (1 - \lambda)(z - x)f(c_2) \quad (14)$$

Adding, dividing by $z - x$ and using (10) and (9) we get

$$\begin{aligned} \frac{I}{z - x} &= \lambda f(c_1) + (1 - \lambda)f(c_2) \\ &\leq \lambda f(c_2) + (1 - \lambda)f(c_2) \\ &= f(c_2) \\ &= \frac{I_2}{z - y} \end{aligned} \quad (15)$$

But that's exactly what is asked. ■

An excellent problem which tests the ability to recast what is to be proved in such a way that the truth of it becomes intuitively obvious. The problem also has a discrete analogue. Suppose $\{a_n\}$ is a monotonically increasing sequence of real numbers. For each n , let b_n be the A.M. of the first n terms. Then for integers p, q , b_{p+q} lies between the average of the first p terms and that of the last q terms. Q.40 of Paper 2 of JEE 2016 was also a problem of a similar spirit.

- Q. 8.** Consider $n(> 1)$ lotus leaves placed around a circle. A frog jumps from one leaf to another in the following manner. It starts from some selected leaf. From there, it skips exactly one leaf in the clockwise direction and jumps to the next one. Then it skips exactly two leaves in the clockwise direction and jumps to the next one. Then it skips three leaves again in the clockwise direction and jumps to the next one, and so on. Notice that the frog may visit the same leaf more than once. Suppose it turns out that if the frog continues this way, then all the leaves are visited by the frog sometime or the other. Show that n *cannot* be odd.

Solution and Comments: Number the leaves from 1 to n . When the frog completes one turn of the pond, it will visit the same leaves (but the gaps will widen). To represent the motion of the frog in a simple manner, let us give the numbers $n + 1$ to $2n$ to the leaves in the second round, $2n + 1$ to $3n$ in the third and so on. This way the positions occupied by the frog will always be represented by a strictly increasing sequence.

Put differently, we number the leaves by all positive integers with the understanding that two integers p and q (with $p < q$) represent the same leaf if and only if the integer $q - p$ is a multiple of n . (This relationship between p and q is called **congruence modulo n** and written as $p \equiv q \pmod{n}$. It is an equivalence relation with n distinct equivalence classes each having a unique representative from 0 to $n - 1$, and in essence we are denoting the leaves with these congruence classes. A familiar example of congruence classes is given by an ordinary watch where $n = 12$. If a clock is showing 2:00 then after 17 hours it will show 7:00 because the congruence class of $2 + 17$, i.e. of 19 modulo 12 is represented by 7. Note that we are treating 12:00 as 0:00 here. The congruence classes are also called **residue classes modulo n** because for any integer m , we can write $m = qn + r$ for some integers q (for quotient) and r (for remainder or residue). Note that even if m is negative, r is to vary between 0 and $n - 1$. Thus the congruence class of -26 modulo 12 is not 2 but 10 as we see by writing $-26 = -3 \times 12 + 10$.)

With this understanding, the successive positions occupied by the frog are numbered as

$$1, 3, 6, 10, 15, 21, 28, 36, 45, 55, 66, 78, 91, 105, 120, 136, 153 \dots \quad (1)$$

Depending on what n is, the actual positions will be represented by the respective residue classes modulo n . For example, when $n = 5$, the sequence, say $\{b_k\}$, of the corresponding residue classes modulo 5 will be

$$1, 3, 1, 0, 0, 1, 3, 1, 0, 0, 1, 3, 1, 0, 0, \dots \quad (2)$$

We note that at least in this case the residue classes recur in cycles of length 5. Also note that the residue classes 2 and 4 do not appear in this sequence. Our problem is to show that for every odd n , some residue class will be missing. Before tackling it, let us carry out the experiment with another odd integer, say 9 and also with an even integer, say 8.

For $n = 9$, the sequence $\{b_k\}$ of residue classes modulo 9 of the terms in (1) is

$$1, 3, 6, 1, 6, 3, 1, 0, 0, 1, 3, 6, 1, 6, 3, 1, 0, 0, \dots \quad (3)$$

Again we notice that the residue classes recur in cycles of length 9. Moreover many residue classes are not represented.

For $n = 8$, the picture is

$$1, 3, 6, 2, 7, 5, 4, 4, 5, 7, 2, 4, 3, 1, 0, 0, 1, 3, 6, \dots \quad (4)$$

This time we see that the residue classes recur in a cycle of length 16 and not 8. Also every residue class from 0 to 7 is represented.

These experiments suggest that for odd n the residue classes recur in a cycle of length n while for an even n they do so in a cycle of length $2n$. Our interest is only in the former and so we confine to it.

Now, coming to the core of the problem, the general formula for the k -th term of the sequence (1) is very well known as it is the sum of the first k positive integers.

$$a_k = \frac{k(k+1)}{2} \quad (5)$$

which can be easily proved by induction on k because from the data we have that $a_{k+1} = a_k + k + 1$. (Usually, this is the first example of mathematical induction taught in schools. The numbers occurring in this sequence are popularly called **triangular numbers** because a_k is

the number of dots in a triangular arrangement with k rows with the first row containing one dot, the second one 2, the third one 3 and so on.)

Coming to the problem, we are given, that for every $r = 1, 2, \dots, n$, the r -th leaf is visited some time, i.e. there exists some k such that $a_k \equiv r \pmod n$. And we are asked to prove that n cannot be odd. We prove the contrapositive. That is, we assume that n is odd, say $n = 2m + 1$ and show that some residue class is missed when in the sequence $\{b_k\}$ where b_k is the residue class of $a_k = \frac{k(k+1)}{2}$.

We begin by showing that the residue classes repeat in a cycle of length n , This is equivalent to showing that

$$a_{k+n} \equiv a_k \pmod n \quad (6)$$

for every k and is best done by a direct computation. For, we have

$$\begin{aligned} a_{k+n} - a_k &= \frac{(k+n)(k+n+1) - k(k+1)}{2} \\ &= \frac{n(n+2k+1)}{2} \end{aligned} \quad (7)$$

When n is odd, $n+2k+1$ is even and so the R.H.S. is a multiple of n . So, $b_{k+n} = b_k$. (We also see that if n is even, then this does not happen. However, a similar computation will show that $b_{k+2n} = b_k$ which means that the residue classes recur in a cycle of length $2n$ which we already observed for $n = 8$. Of course, what happens for an even n is not a part of the problem. But when you are not hard pressed for time it is not a bad idea to venture a little beyond what is asked.)

We are not done yet. We have to show that if n is odd then at least one residue class is not represented in the sequence $\{b_k\}$. When we are dealing with a sequence whose terms recur in a cycle of length n , any block of n consecutive terms is sufficient to determine the sequence. In (2) and (3), which correspond to the cases $n = 5$ and $n = 9$ respectively, we notice that there are two consecutive 0's. We choose any block which contains two consecutive zeros. Now, if in a block of length n , two residue classes are equal, it automatically means that some residue class is missing because in all there are n residue classes and only n places to accommodate them.

Thus our problem is reduced to showing that if n is odd then the sequence $\{b_k\}$ contains two successive zeros. For this we let $n = 2m + 1$. Then

$$a_{2m} = \frac{2m(2m + 1)}{2} = mn \quad (8)$$

which is a multiple of n . So the residue class b_{2m} is 0. Next,

$$a_{2m+1} = \frac{(2m + 1)(2m + 2)}{2} = n(m + 1) \quad (9)$$

which is also a multiple of n and so b_{2m+1} is also 0. Thus two consecutive residue classes are 0 and as shown before this completes the solution. ■

The problem only asks us to show that *some* residue class modulo n is missing if n is odd. It does not ask us to identify the missing residue classes. From (2) we see that when $n = 5$, the residue classes 2 and 4 are missing, while from (7) we see that for $n = 9$, the residue classes 2, 4, 5, 7 and 8 are missing. Guessing what happens for a general odd n and also exploring whether all residue classes are represented when n is even are left as fun time activities. Do more experiments, recognise some patterns, make a tentative guess and try to prove it. That's the name of the game!

There is a related but a lot more difficult problem where once a frog jumps from some leaf it drowns because of the reaction and is not available again to jump onto. See Exercise (24.77).

Although the problem is simple once the essential idea strikes, it takes some experimentation to hit this idea. Such a problem, although not mathematically challenging, deserves to be commended from a pedagogical point of view. The lotus pond setting is also imaginative and different from the usual settings where guests are seated around a table and a ball is passed around in some party game. It reminds one of *Leelavati*, the classic Sanskrit treatise on elementary mathematics by the great Bhaskaracharya.

CONCLUDING REMARKS

As with the commentary on ISI B. Sc. Entrance test 2016, the present commentary was motivated by a search for better problems (as compared with JEE, which is also an examination given at the HSC level). And surely both the papers of 2014 contain a good number of imaginatively designed problems. It is remarkable that this was done even when the syllabus of ISI Entrance Test is much less extensive than that of JEE. In fact, this might as well be a blessing in disguise for the ISI paper setters. They can focus only on a few topics, such as combinatorics, algebra, number theory and geometry which are more fertile sources of problems which test qualities rather than the extent of a candidate's knowledge. The JEE paper setters, on the other hand, are under a perceived compulsion to include at least one question on each of the areas such as probability, vectors, calculus and differential equations where it is getting increasingly difficult to come with really novel problems.

Not that all the problems in the ISI 2014 are novel. Many are such whose likes have been asked in JEE or comparable tests. These include Q.3 (solving a trigonometric equation), Q.4 (about counting six digit numbers of a type), Q.8 (finding the ratio of the areas of two pentagons), Q.10 (distributing chocolates), Q.13 (behaviour of a cubic function), Q.14 (closest point on a locus), Q.19 (isosceles triangles with vertices from an 18-gon), Q. 20 (about existence of a limit of $\frac{0}{0}$ form), Q.23 (increasing/decreasing behaviour of a given function), Q.24 (factoring 1000 into three factors) and Q.25 (evaluation of a limit of the 1^∞ type). Q.5 (evaluating a definite integral using a well known trick) can also be added to this list. But the solution based on an unusual trick tailored for the problem makes it a good problem.

Because of the limited scope of the ISI syllabus, certain areas which were popular in the JEE way back when it also had a limited syllabus (specifically, before 1978) but which now occur rarely in JEE so as to make room for other areas, have made an appearance in ISI 2014. These are, Q.2 (algebra of logarithms), Q.6 (finding the path of a reflected light ray), and Q.17 (points of intersections of the graphs of a function and its inverse function). Similarly, Q.16 (construction of sets starting from the empty set) and Q.28 (about inverses of subsets under a function) represent pure set theory which is now almost non-entity in JEE.

Especially welcome is a revival of geometry, either directly as in Q.11 (about a square with two vertices on a circle and two on a tangent), Q.22

(about a cyclic trapezium) and Q.30 (based on concurrency of the angle bisectors of a triangle) or through complex numbers such as Q.9 (about the circumcentre of a triangle) and Q.26 (about maximising $|z|$ subject to some constraint). As pointed out in the comments on the respective questions, the first two out of these could have been converted to problems of geometric constructions and asked in Paper 2. This most ingenious aspect of geometry is long dead in JEE because, like proofs by induction, it is inherently unsuitable for multiple choice questions. It is time ISI revives it through Paper 2.

Also interesting are problems on number theory, viz. Q.12 (about the largest power of 2 dividing a product of five distinct numbers), Q.18 (about the longest block of integers relatively prime to 374) and Q.15 (about the highest power of 3 dividing the product of the members of a Pythagorean triple of integers). The first two are imaginative problems requiring little more than factorisation.

Q.21 is an excellent problem about finding the area of a certain region without doing any integration and properly belongs to Paper 1. Q.27 is an unusual problem about the minimum of the absolute value of the sum of six trigonometric functions. It belongs to Paper 1 if the idea is merely to reward the ability to think of a tricky substitution. But his trick is likely to elude even the good students unlike the key idea in Q.21. So, it would have been better to ask Q.27 as a full length question where some partial credit can be given to the sincere mortals. Similarly Q.29 (about the minimisation of the area between the graph of a function and a tangent to it) properly belongs to Paper 2 because the justification is vital.

Q.1 is an imaginative application of the A.M.-G.M. inequality while Q.7, although superficially a problem on the integral part of a number, requires its fractional part too in the solution.

Over-all, Paper 1 was a mixed bag. This is to be expected because it is no joke to design 30 good problems as MCQ's.

In Paper 2, on the other hand, the paper setters are free from the constraints of MCQ's and so expectations are high. Simple, single idea questions hardly belong to paper 2. Q.4 (about a sum of consecutive integers being a perfect square) is an example of such a disappointing question. Also, the unnecessarily strong hypothesis of Q.5 obscures its real crux. Q.2 (about the points of intersection of the graph of a fourth degree polynomial with the x -axis) also becomes trivial when reduced to a problem about its roots. This reduction is too obvious to make the problem exciting. One cannot help

wishing that these problems were replaced by some good questions involving geometric constructions, combinatorial identities and proofs by induction which are among the prime casualties of the MCQ format.

But all the remaining five questions in Paper 2 are very good. The last one, about the positions occupied by a frog jumping on lotus leaves in a circle, is a good test of the ability to do some experimentation before getting the key idea.

The over-all experience, like that for the ISI 2016 papers, was that writing this commentary was more enjoyable than writing the JEE Advanced commentary in recent years.